RESEARCH ARTICLE SUMMARY

BIOLOGICAL IMAGING

Achieving optical transparency in live animals with absorbing molecules

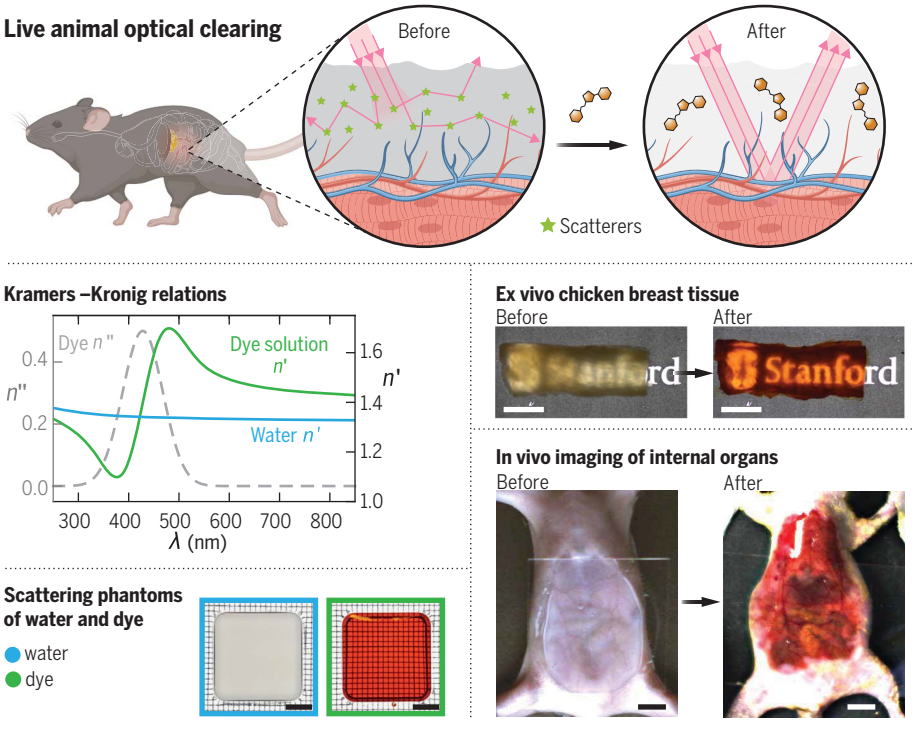
Zihao Ou, Yi-Shiou Duh, Nicholas J. Rommelfanger, Carl H. C. Keck, Shan Jiang, Kenneth Brinson Jr., Su Zhao, Elizabeth L. Schmidt, Xiang Wu, Fan Yang, Betty Cai, Han Cui, Wei Qi, Shifu Wu, Adarsh Tantry, Richard Roth, Jun Ding, Xiaoke Chen, Julia A. Kaltschmidt, Mark L. Brongersma*, Guosong Hong*

INTRODUCTION: A challenge in trying to image biological matter is that its complex structure causes opacity because of unwanted light scattering. This scattering results from refractive index mismatches among the components of biological tissues, limiting the penetration depth of optical imaging. The desire to see inside biological tissue and uncover the fundamental processes of life has spurred extensive research into deep-tissue optical imaging methods, such as two-photon microscopy, near-infrared-II fluorescence imaging, and optical tissue clearing. However, these methods either lack sufficient penetration depth and resolution or are unsuitable for living animals. Therefore, the ability to achieve optical transparency in live animals holds promise for transforming many optical imaging techniques.

RATIONALE: We hypothesized that strongly absorbing molecules can achieve optical transparency in live biological tissues. By applying the Lorentz oscillator model for the dielectric properties of tissue components and absorbing molecules, we predicted that dye molecules with sharp absorption resonances in the near-ultraviolet spectrum (300 to 400 nm) and blue region of the visible spectrum (400 to 500 nm) are effective in raising the real part of the refractive index of the aqueous medium at longer wavelengths when dissolved in water, which is in agreement with the Kramers-Kronig relations. As a result, water-soluble dyes can effectively reduce the RI contrast between water and lipids, leading to optical transparency of live biological tissues.

RESULTS: Following our theory, we found that an aqueous solution of a common food color approved by the US Food and Drug Administration, tartrazine, has the effect of reversibly making the skin, muscle, and connective tissues transparent in live rodents. We conducted experiments in both tissue-mimicking scattering hydrogels and ex vivo biological tissues. These tests confirmed the mechanism underlying our observations and showcased the achievable spatial resolution down to the micrometer level through millimeters of scattering medium once transparency is attained. By using absorbing dye molecules, we can transform the typically opaque abdomen of a live mouse into a transparent medium. This "transparent abdomen" allows for direct visualization of fluorescent protein–labeled enteric neurons, capturing their movements that mirror the underlying gut motility in live mice. This enabled us to generate time-evolving maps that depict mouse gut motility and the diversity of movement patterns. To demonstrate the generalizability of this approach, we also applied dye solutions topically to the scalp of a mouse head for visualizing cerebral blood vessels and to the mouse hindlimb for high-resolution microscopic imaging of muscle sarcomeres.

CONCLUSION: Overall, we report the counterintuitive observation that strongly absorbing molecules can achieve optical transparency in live animals. The Lorentz oscillator model, which underlies this unusual observation, predicts that molecules with low resonance frequencies (long absorption wavelengths), sharp absorption peaks, and rich delocalized electrons are more effective candidates at raising the refractive index of the aqueous medium than are conventional optical clearing agents. Our approach also presents opportunities for visualizing the structure, activity, and functions of deep-seated tissues and organs without the need for surgical removal or the replacement of overlying tissues with transparent windows. Some limitations remain for this method, including reduced but not eliminated scattering owing to the challenge of matching refractive indices in heterogeneous tissues and achievable penetration depth depending on the diffusion of absorbing molecules. ■



Achieving optical transparency in live mice with absorbing dye molecules. Strongly absorbing molecules dissolved in water can modify the RI of the aqueous medium through the Kramers-Kronig relations to match that of lipids. This approach can render various samples transparent, including scattering phantoms, chicken breast tissue, and live mouse body for visualizing a wide range of deep-seated structures and activities. Scale bars, 5 mm. [The schematic was prepared using BioRender.com]

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Cite this article as Z. Ou *et al.*, *Science* **385**, eadm6869 (2024). DOI: 10.1126/science.adm6869

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https://doi.org/10.1126/science.adm6869

RESEARCH ARTICLE

BIOLOGICAL IMAGING

Achieving optical transparency in live animals with absorbing molecules

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Optical imaging plays a central role in biology and medicine but is hindered by light scattering in live tissue. We report the counterintuitive observation that strongly absorbing molecules can achieve optical transparency in live animals. We explored the physics behind this observation and found that when strongly absorbing molecules dissolve in water, they can modify the refractive index of the aqueous medium through the Kramers-Kronig relations to match that of high-index tissue components such as lipids. We have demonstrated that our straightforward approach can reversibly render a live mouse body transparent to allow visualization of a wide range of deep-seated structures and activities. This work suggests that the search for high-performance optical clearing agents should focus on strongly absorbing molecules.

The complex structure of biological matter causes opacity because of unwanted scattering and absorption of light, which limits the penetration depth of optical imaging (*1*, *2*). In most tissues, the scattering coefficient is 10 to 1000 times larger than the absorption coefficient (*2*, *3*); thus, scattering processes can severely limit the imaging depth and spatial resolution in conventional microscopy (*4*). If one could achieve substantial reductions in light scattering, this could lead to substantial enhancement of brightfield, fluorescence, nonlinear, and super-resolution imaging techniques.

Light scattering in tissue originates from the difference between low refractive index (RI) aqueous-based components (such as the interstitial fluid and cytosol) and high RI lipid- and protein-based components (such as the plasma membrane, myelin, and myofibrils). Existing methods to reduce this difference in RI usually replace water with high-RI chemicals or remove lipids to yield an all-aqueous environment (*5*–*8*). Despite their success, these approaches are seldom used in live tissues because they involve the use of toxic substances [such as tetrahydrofuran (*8*) and acrylamide (*6*)] and removal of molecules vital to sustaining life (such as water and lipids).

We report on the counterintuitive observation that strongly absorbing molecules can achieve optical transparency in live biological tissues. Specifically, we found that an aqueous solution of a common food color approved by the US Food and Drug Administration, tartrazine, has the effect of reversibly making the skin, muscle, and connective tissues transparent in live rodents. We explored the physical principles behind this phenomenon. Applying the Lorentz model for the dielectric properties of materials, we modeled tartrazine, along with other absorbing molecules, as a Lorentzian oscillator (*9*). Our theoretical model revealed that Lorentzian oscillators with sharp absorption resonances in the visible spectrum (400 to 750 nm) are very effective in raising the real part of the RI in the red and infrared parts of the spectrum when dissolved in water, which is in agreement with the Kramers-Kronig relations (*10*, *11*). As a result, water-soluble dyes in the visible spectrum can effectively reduce the RI contrast between water and lipids, leading to optical transparency of live biological tissues. We validated the efficacy of inducing optical transparency by use of visible dyes in highly scattering tissue-mimicking phantoms and dissected chicken breast. This approach is fundamentally different from previous reports on absorption-enhanced resolution (*12*, *13*). Specifically, the optical transparency observed in our study does not arise from the absorption of scattered photons by dye molecules but rather from reduced scattering events in the tissue.

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Absorbing molecules make skin and muscle transparent in live mice

Tartrazine is a yellow-to-orange, water-soluble mono-azo compound used as a common synthetic pigment in food manufacturing owing to its high absorbance in the blue region of the spectrum (*14*). It is a hygroscopic compound with a high solubility in water and minimal reactivity. The European Food Safety Authority (EFSA) has concluded that this molecule exhibits no adverse effects or systemic toxicity at doses up to 2 g per kilogram of body weight (*14*). As a result, it is widely used as an artificial food dye in products such as flavored chips (for example, Doritos). When dissolved in water, a tartrazine solution displays its characteristic yellow-to-orange color (Fig. 1A), hence its common name "FD&C Yellow 5." The ultraviolet-visible (UV-vis) absorption spectrum of a tartrazine solution exhibits two major peaks located at 257 and 428 nm but with minimal absorption beyond ~600 nm (fig. S1, A and B). We verified the chemical composition of tartrazine used in this study with mass spectrometry (fig. S2).

We first demonstrated the ability of a dye solution to eliminate scattering from an otherwise opaque suspension containing colloidal silica (RI = 1.43) particles dispersed in an aqueous solution. Specifically, 0.6 M tartrazine solution achieved complete optical transparency in the red region of the visible spectrum for this originally opaque medium. By contrast, conventional optical clearing agents (OCAs) such as glycerol—which has been demonstrated to moderately decrease scattering in vivo (*15*, *16*)—cannot attain the same transparency in the scattering medium at the same concentration (Fig. 1, A and B). We hypothesized that absorbing dyes could achieve optical transparency in live tissues by suppressing the scattering inherent to biological structures.

We applied tartrazine solutions topically to the scalp of a shaved, live mouse head and used laser speckle contrast imaging (LSCI) to visualize cerebral blood vessels in the head (Fig. 1C). LSCI usually requires removal of the scalp because of its opacity (*17*), and imaging of the shaved mouse head did not reveal any interesting features of cerebral vessels (Fig. 1, D and E). By contrast, after applying the tartrazine solution, LSCI reveals structures representative of cerebral vessels (Fig. 1F and fig. S3 and S4). To confirm the location of these vessels in the brain, we performed LSCI after removing the scalp, identifying similar structures (Fig. 1F, inset). Besides tartrazine, a few other absorbing molecules demonstrated a similar degree of scalp transparency (Materials and methods and fig. S5). Topically applied dye molecules can be removed from the scalp by rinsing with water, effectively reversing the transparency effect in a repeatable manner (fig. S6). In addition to topical application,

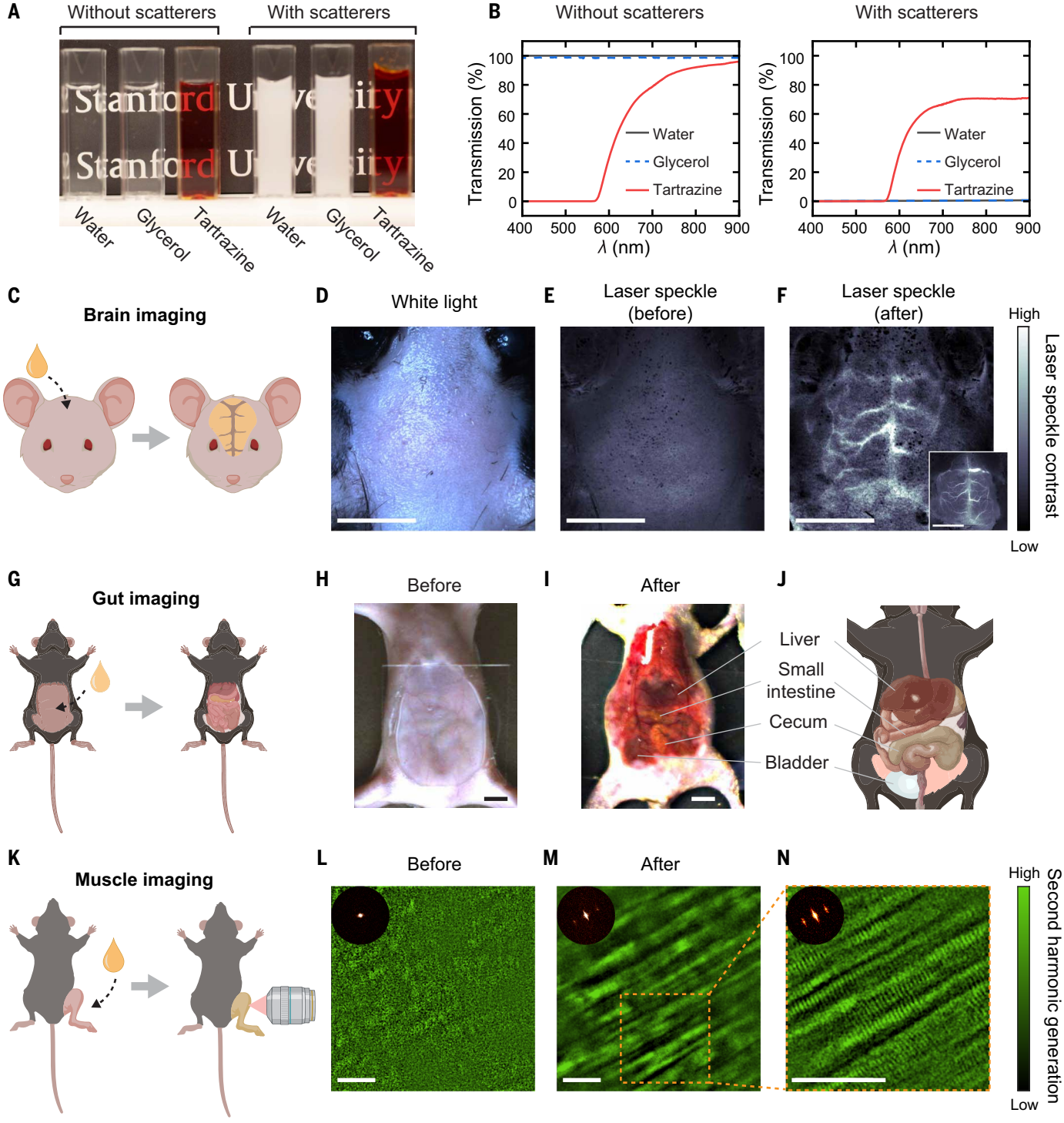
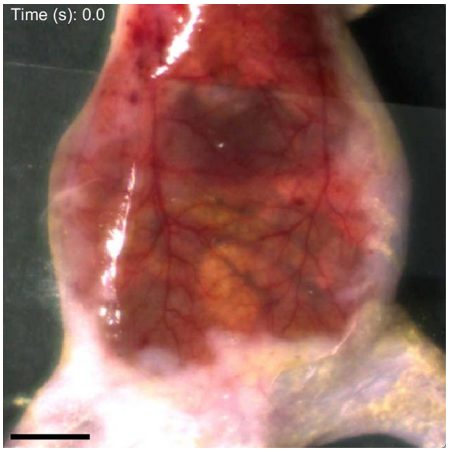
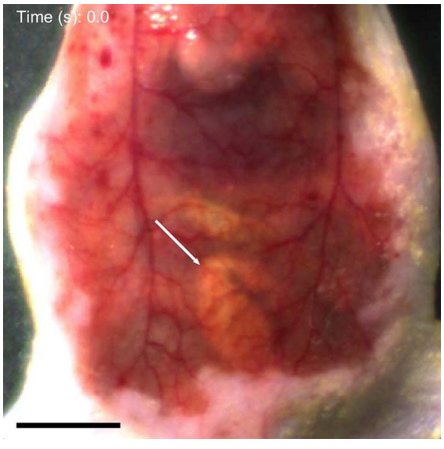


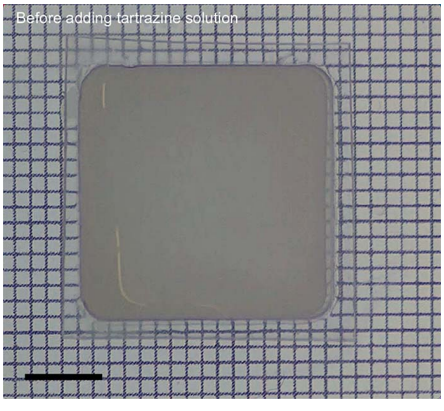
Fig. 1. Absorbing molecules make live mice optically transparent. (**A**) Photograph showing increased optical transparency of a scattering medium after introducing absorbing molecules into it. From left to right, the first three 1-cm cuvettes include pure water, a 0.6 M glycerol solution, and a 0.6 M tartrazine solution, respectively. The other three cuvettes contain colloidal silica particles, which serve as scatterers, dispersed in hydrogels prepared with the three previously mentioned solutions. (**B**) Transmission spectra of samples shown in (A). (**C**) Schematic illustrating the process of making the scalp transparent. (**D**) Photograph of the depilated mouse scalp before topical application. (**E** and **F**) Laser speckle contrast images of the mouse head (E) before and (F) after topical application. (Inset) The laser speckle contrast image of the same mouse head after the scalp is removed. Scale bars (D) to (F), 5 mm. (**G**) Schematic illustrating the process of making the abdominal wall transparent. (**H** to **J**) White-light photographs of the mouse abdomen (H) before and (I) after topical application, with (J) major abdominal organs identified. Scale bars (H) and (I), 5 mm. (**K**) Schematic illustrating the process of making the hindlimb transparent. (**L** to **N**) SHG images of the mouse tibialis anterior muscle (L) before and [(M) and (N)] after topical application. (N) A digital zoom of the red dashed box in (M). (Insets) The fast Fourier transform of each image. Scale bars (L) to (N), 50 μm. [Schematics were created with BioRender.com]



Movie 1. Real-time video demonstrating the optical transparency in the mouse abdomen, enabling the visualization of the abdominal organs. Scale bar, 5 mm.



Movie 2. Real-time video showing the motility of the intestines through the transparent abdomen in a live mouse. Scale bar, 5 mm. The white arrow indicates peristalsis of the cecum.



Movie 3. Real-time imaging showing the dynamic process of a scattering phantom transitioning from opacity to transparency after immersion in a 0.6 M solution of tartrazine. Scale bar, 5 mm.

injecting dye molecules into the scalp produces similar LSCI images (fig. S7), offering an alternative method for achieving tissue transparency in thicker skins.

We next sought to demonstrate the ability of absorbing molecules to achieve tissue transparency in the mouse abdomen. Specifically, when a tartrazine solution was topically applied to the abdominal skin of a live mouse under anesthesia and gently massaged on the skin (Fig. 1, G and H), the abdominal skin not only darkened in color but also became more transparent in the red window (fig. S8). This transparency effect can be readily visualized with the naked eye and does not require any specialized imaging equipment. A few absorbing molecules have been proven to render the mouse abdomen transparent (Materials and methods section). The transparent abdomen allowed us to directly observe internal organs, including the liver, small intestine, cecum, and bladder (Fig. 1, I and J). Moreover, we could also discern their movements, such as peristalsis, as well as motions synchronized with the heartbeat and breathing (Movies 1 and 2). The achieved abdominal transparency can be reversed by rinsing and massaging the skin with water (fig. S8).

We sought to demonstrate microscopic imaging in deep tissue with topically applied tartrazine solutions. We chose sarcomeres, basic contractile units of skeletal and cardiac muscles, in the mouse hindlimb as our targets for imaging because of their central roles in muscle functions and implications in neuromuscular diseases (such as spinal muscular atrophy) (*18*). Existing in vivo sarcomere imaging methods require the invasive implantation of microendoscopes inside the muscle tissue to allow imaging of the second-harmonic generation (SHG) signals from myosin rods in the sarcomere, in which case the field of view is limited by the location of the implant (*19*). In our experiments, we topically applied a tartrazine solution on the hindlimb of a mouse (Fig. 1K and fig. S9). Although direct SHG imaging [excitation (Ex), 1040 nm; emission (Em), 520 nm] through the intact skin revealed indistinct microscopic features at a depth of ~220 μm (Fig. 1L), skin in contact with tartrazine became visibly transparent, allowing us to identify the periodic structures of sarcomeres by collecting the SHG signals at the same depth (Fig. 1, M and N, and figs. S10, A to F, S11, and S12). Specifically, the images in fig. S11 were taken at higher excitation numerical apertures (NAs) of 0.5 and 0.8, confirming that our approach is compatible with high-resolution microscopy. A characteristic periodicity of 2.60 ± 0.28 μm (fig. S10, G and H), which corresponds to the length of sarcomeres, was identified in myofibrils through the transparent hindlimb tissue. This average periodicity in the SHG images agrees with the reported sarcomere lengths (*19*), thus confirming the ability of tartrazine to make the tissue transparent and resolve microscopic features in the muscle.

Physical mechanism of optical transparency in live tissues by absorbing molecules

Despite the strong absorption of tartrazine in the UV and blue region of the spectrum, in solution it remains minimally absorbing above 600 nm, even up to a concentration of 0.78 M (fig. S1). The high transmittance in the ≥600 nm spectrum agrees with our observation that tissue transparency is primarily achieved in the red wavelengths. Therefore, we speculated that the observed transparency stems from increased absorption at wavelengths shorter than those exhibiting transparency.

We hypothesized that dye molecules with strong absorption and superior solubility can substantially increase the RI of the aqueous medium at wavelengths immediately longer than the absorption upon dissolution. We confirmed this with the Kramers-Kronig relations that connect the spectral dependences of the real (n') and imaginary (n'') components of the RI of a material through causality (*11*)

$$n'(\lambda) = 1 + \frac{2}{\pi}\mathrm{P.V.}\int_0^\infty n''(\lambda')\left[\lambda'\left(1-\frac{\lambda'^2}{\lambda^2}\right)\right]^{-1} d\lambda' \quad (1)$$

where λ is the wavelength of light and P.V. denotes the Cauchy principal value of the integral. This equation shows that the spectral location, spectral width, and absorption strength of a dye are all parameters that control the achievable changes in $n'(\lambda)$. As a numerical example, substantial changes in the n' of the solution (on the order of 0.1) can be realized in the visible spectrum by dissolving tartrazine molecules with a Gaussian absorption peak centered at 428 nm, a full-width at half-maximum (FWHM) of 94 nm, and a peak absorption coefficient of $\mu_a = 4\pi n''\lambda^{-1} = 0.015\ \mathrm{nm}^{-1}$ (corresponding to a value of $n'' = 0.5$) (Fig. 2A). Consequently, the n' of the aqueous solution increases substantially at $\lambda > 428$ nm, reaching that of lipids and collagen fibers ($n' = 1.43$ to 1.53) (*2*). n' stays higher than that of pure water ($n' = 1.33$), even at wavelengths at which n'' has essentially returned to zero and the absorption is negligible (for example, >800 nm). The achievable changes in n' are proportional to the increase in n'' (fig. S13), thus suggesting that water-soluble dye molecules with high n'' are effective agents to increase n' of the aqueous medium at wavelengths longer than the absorption upon dissolution.

These findings align with reported effects showing that absorbing molecules can increase the RI of a medium when dissolved or dispersed within it. For example, water-soluble food dyes (such as Brilliant Blue, Allura Red, Pyranine, and Quinoline Yellow) have been

shown to substantially alter the RI of aqueous solutions upon dissolution (*20*). Similarly, plant pigments (such as chlorophylls, carotenoids, and anthocyanins) can enhance the RI of commercially available polymers when dispersed within their matrices (*21*). Additionally, hematoxylin and eosin stains are noted for changing the RI of intracellular components (*22*). However, none of these studies have explored the use of absorbing molecules to reduce scattering in their systems, let alone achieve tissue transparency in vivo (Fig. 1). Apart from these "RI engineering" studies, observations have shown that naturally occurring, strongly absorbing pigments in apple skin can influence light scattering by altering the RI (*23*). This study does not actively use apple skin pigments to alter optical properties but instead highlights how nature uses the Kramers-Kronig relations.

Optimizing the spectral properties of a dye molecule would enable optical transparency at a concentration that is both low and physiologically tolerable. To this end, we simulated optical behavior of the dye molecule using the Lorentz model with a single optical resonance (*9*). We show that the multioscillator Lorentz model can be reduced to a single-oscillator model for molecules with multiple absorption peaks (supplementary text 1). With this model, we can directly link the relative permittivity ε_r (and complex RI $n = \sqrt{\varepsilon_r}$) of the OCA to its plasma frequency ω_p and damping constant γ

$$\varepsilon_r = \varepsilon_\infty + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega} \quad (2)$$

Here, ε_∞ is the high-frequency dielectric constant, γ is approximately equal to the FWHM of the resonance peak. Choosing an operating frequency ω in the vicinity of a resonance at $\omega = \omega_0 - \Delta\omega$ simplifies the expression to

$$\varepsilon_r \approx \varepsilon_\infty + \frac{\omega_p^2}{\omega_0} \cdot \frac{1}{2\Delta\omega - i\gamma} \quad (3)$$

Equation 3 reveals a number of aspects that govern the efficacy of an absorbing molecule in raising the n' of the medium it is dissolved in. First, given the ${\omega_0}^{-1}$ scaling, absorbing molecules with low resonance frequencies (small ω_0) are more effective in raising the n' of the medium. It is also strategic to choose a resonance frequency that lies just above the frequency used for imaging (a small positive-valued $\Delta\omega$). This ensures that the maximum increase in the permittivity is obtained owing to the resonance without a detrimental amount of absorption. Because optical imaging is usually performed in the visible spectrum, these requirements suggest that dyes with absorption peaks in the blue region of the spectrum make more effective OCAs than do those with absorption peaks in the near-UV (NUV; 300 to 400 nm) and shorter wavelengths. Second, the absorbing molecule should have a low damping constant γ, which is reflected in its absorption spectrum as a narrow peak with a small FWHM. Equation 3 predicts that a dye molecule with sharp absorption peaks acts as a more intense Lorentz oscillator with a larger ε_r value near its resonance. We also wanted to avoid substantial attenuation due to absorption at the imaging wavelength. For the strongly absorbing dye considered in this work, we thus stayed a bit away from the resonance. Empirically, we found that the dye molecules we used (Fig. 1) were operated at $\omega = \omega_0 - \gamma$ to achieve sufficient transparency.

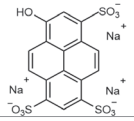
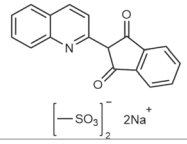
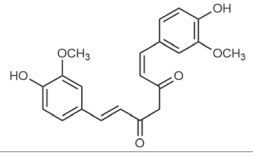
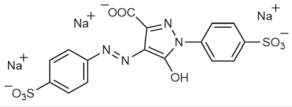
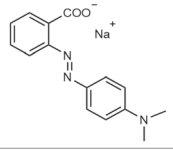
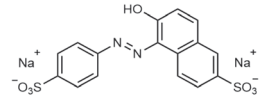
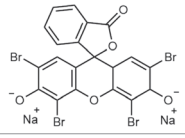
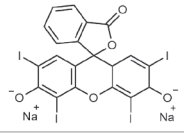
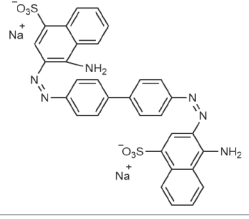
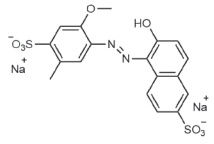
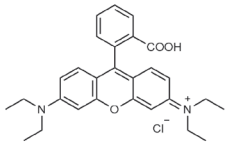
The Kramers-Kronig relations (Eq. 1) and the Lorentz model (Eqs. 2 and 3) predicted that as the resonance absorption wavelength of absorbing molecules increased, they would become more effective in raising n' of the medium. To illustrate this point, we modeled three Lorentz oscillators with their resonances at 100, 250, and 400 nm (Fig. 2, B and C). The Lorentz resonator at 100 nm exhibited weak absorption in n'', leading to a small increase in n' in the visible spectrum. By contrast, the Lorentz resonator at 400 nm was predicted to be a stronger absorber with a higher peak n'', thus resulting in a more efficient increase in n' at longer wavelengths. This theoretical prediction was validated experimentally, as evidenced by the small molar absorption coefficient α of 0.012 M^{-1} at 84 nm of the common OCA glycerol, corresponding to a small molar $\Delta n'$ coefficient β (defined as the change in n' of the medium induced per molar concentration of dissolved OCA) (a rigorous definition of β is provided in Materials and methods, and the derivation of β is available in supplementary text 2) of 0.0129 M^{-1} in the visible spectrum upon dissolution in water (Fig. 2, D and E, and Table 1). This finding reveals that conventional OCAs such as glycerol raise the RI of tissue because they are themselves absorbing molecules in the short-wavelength, extreme-UV (EUV) spectrum (usually <150 nm). However, because they absorb at extremely short wavelengths (likely intended to be completely transparent in the visible spectrum), conventional OCAs are not strong absorbers and thus inefficient at raising the n' of the medium. For example, with a small β of 0.0129 M^{-1} for glycerol, one must use glycerol at a concentration of 11.6 M (almost pure glycerol) to raise the baseline n' of water to match that of lipids. This requirement, which is deeply rooted in physics, explains why glycerol necessarily results in dehydration and shrinking of tissue when applied as an OCA.

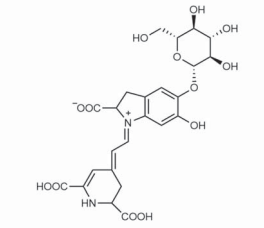
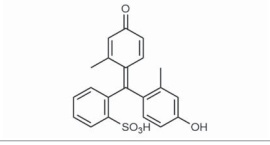
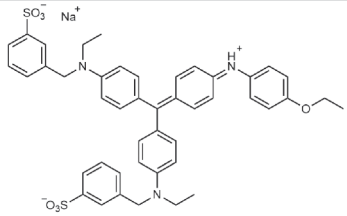
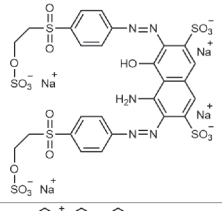
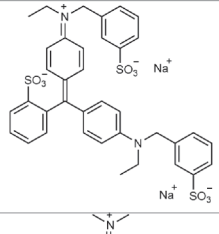
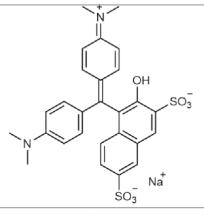
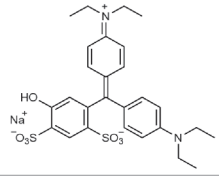
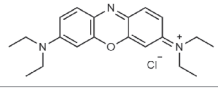
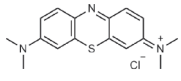
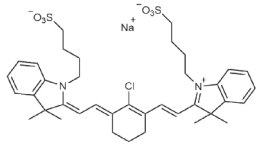
This theoretical prediction also agrees with the α and β spectra of tartrazine (Fig. 2, F to I, Table 1, and fig. S14), with a high absorption coefficient of $(2.04 \pm 0.02) \times 10^4$ M^{-1} cm^{-1} at its peak absorption of 428 nm (Fig. 2, D and E, and fig. S1C), while becoming completely transmissive beyond 600 nm (Fig. 2J). Tartrazine is more than 15 times more efficient than glycerol in raising the n' of water (Fig. 2E) and two to three orders of magnitude more efficient than other commonly used OCAs (Table 1). We further confirmed this dependence of the RI increase on the resonance frequency by plotting α and β of 23 absorbing molecules versus their peak absorption from 84 to 800 nm, with stronger absorption and a more efficient n' increase at longer wavelengths (Fig. 2, K and L, and Table 2). These 23 molecules include 21 absorbing molecules in the visible spectrum, along with the common OCAs glycerol and antipyrine, the latter of which absorbs in the short-wavelength UV spectrum. Antipyrine represents one of the best performing n'-matching agents after screening more than 1600 chemicals (*24*); however, it has a much lower β than that of the visible-absorbing molecules, confirming the wavelength dependence of the achievable n' increase imposed by the Lorentz oscillator model.

Table 1. Efficiency of absorbing dye molecules in achieving optical transparency compared with that of conventional OCAs.

Chemical	Wavelength (nm)	Molar n' change (M^{-1})	Reference and notes
Tartrazine	500	0.198	This work
	550	0.158	
	600	0.143	
	800	0.130	
Dye-21 (Table 2)	798	1.27	This work
Antipyrine	600	0.037 ± 0.003	This work
Tetrahydrofuran	600	0.0032 ± 0.0007	This work
Glucose	589	0.0258	(*2*)
Sucrose	589.6	0.0462	(*47*)
Sodium chloride	589	0.00094	(*2*)
Propylene glycol	589.6	0.0084	(*47*)
Glycerol	589.6	0.0129	(*47*)

Table 2. Absorbing dye molecules used in this study. <α> and <β> are the molar absorption coefficient and molar $\Delta n'$ coefficient averaged over the 500 to 700 nm range.

DYE INDEX	IUPAC NAME	CHEMICAL STRUCTURE	<α> (M^{-1})	<β> (M^{-1})	SOURCE
1	Trisodium 8-hydroxy-1,3,6-pyrenetrisulfonate		1.1×10^{-4}	0.041	FastColors
2	Sodium 1*H*-indene-1,3(2*H*)-dione- 2-(2-quinolinyl)sulfonate		0.0055	0.071	FastColors
3	1,7-bis(4-hydroxy-3-methoxyphenyl)-(1*E*,6*E*)-1,6-Heptadiene-3,5-dione		5.5×10^{-4}	0.0018	FastColors
4	Sodium 5-hydroxy-1-(4-sulfonatophenyl)-4-[(*E*)-(4-sulfonatophenyl) diazenyl]-1*H*-pyrazole-3-carboxylate		0.0015	0.15	Sigma-Aldrich
5	Sodium 4-dimethylaminoazobenzene-2′-carboxylate ("methyl red")		0.0066	0.080	Sigma-Aldrich
6	Disodium 6-hydroxy-5-[2-(4-sulfophenyl) diazenyl]-2-naphthalenesulfonate		0.028	0.10	Sensient Technologies
7	Disodium spiro[isobenzofuran-1(3*H*), 9′-[9*H*]xanthen]-3-one- 2′,4′,5′,7′-tetrabromo-3′,6′-dioxide		0.12	0.17	FastColors
8	Disodium spiro[isobenzofuran-1(3*H*), 9′-[9*H*]xanthen]-3-one- 2′,4′,5′,7′-tetraiodo-3′,6′-dioxide		0.13	0.16	FastColors
9	Disodium 3,3′-([1,1′-biphenyl]-4,4′-diyl)bis (4-aminonaphthalene-1-sulfonate) ("Congo red")		0.10	0.22	Sigma-Aldrich
10	Disodium 6-hydroxy-5-[(2-methoxy-5-methyl-4-sulfophenyl)azo]-2-naphthalenesulfonate		0.048	0.099	FastColors
11	9-(2-Carboxyphenyl)-6-(diethylamino)-*N*,*N*-diethyl-3*H*-xanthen-3-iminium chloride ("Rhodamine B")		0.20	0.16	Alfa Aesar

DYE INDEX	IUPAC NAME	CHEMICAL STRUCTURE	<α> (M^{-1})	<β> (M^{-1})	SOURCE
12	4-[2-[(2S)-2-carboxy-5-(β-D-glucopyranosyloxy) -2,3-dihydro-6-hydroxy-1*H*-indol-1-yl] ethenyl]-2,3-dihydro-(2S)-2,6-Pyridinedicarboxylic acid		7.9×10^{-4}	9.3×10^{-4}	FastColors
13	4,4′-(1,1-dioxido-3*H*-2,1-benzoxathi- ol-3-ylidene)bis[3-methyl-phenol		0.021	0.057	Matheson Coleman & Bell
14	Sodium benzenemethanaminium *N*-[4-[[4-[(4-ethoxyphenyl)amino]phenyl] [4-[ethyl[(3-sulfophenyl)methyl]amino]phenyl] methylene]-2,5-cyclohexadien-1-ylidene] -*N*-ethyl-3-sulfonate		0.24	0.11	Fisher Scientific
15	Tetrasodium 4-amino-5-hydroxy- 3,6-bis[2-[4-[[2-(sulfooxy)ethyl]sulfonyl] phenyl]diazenyl]-2,7-naphthalenedisulfonate		0.19	0.12	Sigma-Aldrich
16	Disodium benzenemethanaminium *N*-ethyl-*N*-[4-[[4-[ethyl[(3-sulfophenyl) methyl]amino]phenyl](2-sulfophenyl) methylene]-2,5-cyclohexadien-1-ylidene] -3-sulfonate		0.32	0.036	FastColors
17	Sodium 4-[[4-(dimethylamino)phenyl]- (4-dimethyliminiocyclohexa-2,5-dien-1-ylidene) methyl]-3-hydroxy-7-sulfo-naphthalene-2-sulfonate		0.33	0.058	FastColors
18	Sodium 2-[[4-(diethylamino)phenyl]- (4-diethyliminiocyclohexa-2,5-dien-1-ylidene) methyl]-4-hydroxy-5-sulfo-benzenesulfonate		0.30	0.020	FastColors
19	3,7-bis(diethylamino)-phenoxazin-5-ium chloride		0.15	−0.0029	Sigma-Aldrich
20	3,7-bis(dimethylamino)phenazathionium chloride ("methylene blue")		0.27	0.0094	Sigma-Aldrich
21	Sodium 4-[2-[2-[2-chloro-3-[2-[3,3-dimethyl -1-(4-sulfonatobutyl)indol-1-ium-2-yl]vinyl] cyclohex-2-en-1-ylidene]ethylidene]-3,3- dimethyl-indolin-1-yl]butane-1-sulfonate		0.27	−0.32	Sigma-Aldrich

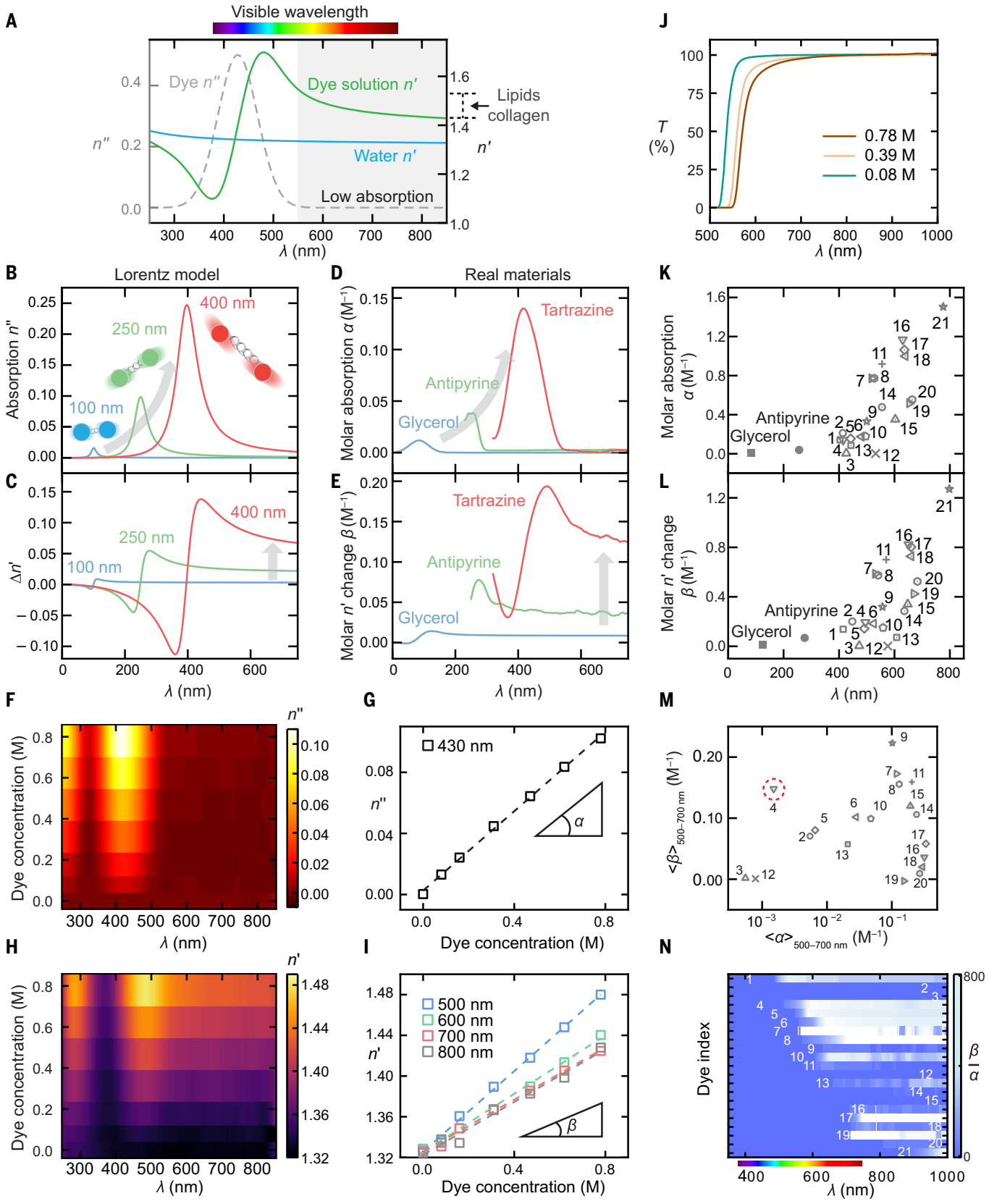


Fig. 2. Physical mechanism for achieving optical transparency with absorbing molecules. (**A**) Numerical simulation showing the modulation of the real index of an aqueous solution (n'; green solid line) by introducing an absorbing molecule with a peak in n'' (gray dashed line) centered at 428 nm. The n' of water (blue solid line) and high-RI cellular components (such as lipids and collagen fibers) are also shown. (**B**) Imaginary part n'' and (**C**) change

in the real part $\Delta n'$ of the RI calculated from a Lorentz oscillator model with resonance frequency ω_0 at 100 nm (blue), 250 nm (green), and 400 nm (red), respectively. (**D**) Molar absorption α and (**E**) molar n' change β of glycerol (blue), antipyrine (green), and tartrazine (red) dissolved in water, respectively. The data for glycerol below 250 nm were obtained from (*51*); the full range ellipsometry data for tartrazine is shown in fig. S14. (**F**) Imaginary part of RI, n'', of tartrazine solutions at different concentrations measured with ellipsometry. (**G**) Dependence of n'' on molar concentrations at 430 nm. Dashed lines indicate linear fitting of the data, from which molar absorption α is extracted. (**H**) Real part of RI, n', of tartrazine solutions at different concentrations measured with ellipsometry. (**I**) Dependence of n' on molar concentrations at 500, 600, 700, and 800 nm. Dashed lines indicate linear fitting of each curve, from which the slopes are extracted to represent the molar n' change, β, at each wavelength. (**J**) Transmittance T of tartrazine solutions with an optical path length of 1 mm measured at different concentrations, showing the transmission window beyond 600 nm despite strong absorption below 500 nm. (**K** and **L**) Wavelength dependence of (K) maximum molar absorption α and (L) maximum molar n' change β for glycerol, antipyrine, and 21 absorbing molecules listed in Table 2. (**M**) The relation between average molar absorption α and average molar n' change β in the visible spectrum for different dye molecules. Tartrazine is highlighted as Dye-4 in the red dotted circle. Dye-21 is not shown because of the large negative β in the wavelength range of interest. (**N**) The ratio between the molar n' change and molar absorption (β/α) for different absorbing molecules. The "white window" indicates a potential transparent spectrum with sufficient $\Delta n'$ and minimal absorption.

Table 3. Comparison between absorbing-molecule–enabled optical transparency and other existing tissue clearing and in vivo optical imaging methods. N.A., not applicable. Abbreviations for the chemicals are THF, tetrahydrofuran; DCM, dichloromethane; BABB, mixture of one volume of 100% benzyl alcohol with two volumes of 100% benzyl benzoate.

	Application	Preservation of endogenous tissue components?	Representative clearing agents	Time to optical transparency	Penetration depth ≥1 mm in tissues?	Resolution down to μm?	Representative reference
Absorbing-molecule-enabled optical transparency	In vivo, topical	Yes	Tartrazine (0.3 to 0.6 M)	Minutes	Yes (diffusion-limited)	Yes	This work
Ex vivo tissue clearing methods							
Hydrophobic method (such as vDISCO)	Ex vivo, soaking	No	THF (12 M), DCM (16 M), and BABB	Hours to days	Yes	Yes	(*30*)
Hydrophilic method (such as CUBIC)	Ex vivo, soaking	No	Urea (4 M) and sucrose (1.5 M)	Days	Yes	Yes	(*7*)
Hydrogel-based method (such as PACT)	Ex vivo, polymerization and soaking	No	Cross-linker, 0.6 M acrylamide RI matching, 1.1 M histodenz	Days	Yes	Yes	(*52*)
In vivo optical imaging methods							
Confocal microscopy	In vivo	Yes	N.A.	N.A.	No	Yes	(*53*)
Light sheet microscopy	In vivo	Yes	N.A.	N.A.	No	Yes	(*54*)
Two-photon microscopy	In vivo	Yes	N.A.	N.A.	Yes	Yes	(*55*)
Three-photon microscopy	In vivo	Yes	N.A.	N.A.	Yes	Yes	(*56*)

The Lorentz oscillator model, along with the Kramers-Kronig relations, predicts that optically absorbing molecules can bring about more substantial and desired changes in the optical properties than conventional RI-matching agents. According to the dependence revealed in Eq. 3 and Fig. 2, K and L, the most efficient agent should have its peak absorption at the longest possible wavelength, which is upper-bounded by the imaging wavelength, to maximize n'' and $\Delta n'$. At the same time, the agent should have a single narrow absorption peak without any additional absorption at the imaging wavelength. In addition to these purely optical requirements, the agents also need to display a high solubility and diffusivity in water as well as excellent biocompatibility to facilitate in vivo application. After sampling 21 candidates (Fig. 2, M and N, Table 2, and fig. S15), tartrazine emerged as the standout candidate that satisfied all requirements for optical transparency in the visible spectrum (Fig. 2N). Specifically, tartrazine has a narrow absorption peak near the short extreme of the visible spectrum, and its solution is minimally absorbing above 600 nm (Fig. 2J and fig. S1). In addition, it exhibits the highest β while maintaining a low α in the visible spectrum (Fig. 2M), as evidenced by a "clear window" that appears at the shortest wavelength in the β/α spectrum for all molecules (Fig. 2N). Emissions down to 500 nm were detectable after achieving optical transparency with tartrazine, despite increased absorption near its resonance peak at 428 nm (fig. S16). This finding suggests that the reduction in scattering outweighs the increase in absorption by tartrazine down to 500 nm.

The dye screening also revealed several other findings. First, 14 dyes other than tartrazine offer similar or even greater potential for achieving tissue transparency at longer wavelengths, with Dye-21 showing the greatest potential in the near-infrared (NIR) spectrum (Fig. 2, L and N). Second, we predicted in vivo optical transparency to be more readily achieved in the NIR spectrum, owing to the exceptionally high values of β attained by resonant absorbers at longer wavelengths. Specifically, Dye-21 exhibited a high β of >1.2 M^{-1} at 800 nm (Fig. 2L), which is ~100 times more efficient than glycerol and similar OCAs (Table 1), as predicted with the Lorentz oscillator model in Eq. 3.

Last, the Lorentz oscillator model also provides theoretical foundation for the long-standing challenge associated with achieving in vivo tissue transparency. Despite recent advances in tissue clearing techniques (*5*), most are effective in ex vivo tissues, with very few applicable in vivo (Table 3) (*25*, *26*). Our theoretical framework, grounded in the Lorentz oscillator model

and Kramers-Kronig relations, offers insights into this challenge. Conventional OCAs such as glycerol (*27*), sucrose (*28*, *29*), and benzyl alcohol/benzyl benzoate (BABB) (*30*, *31*) function as high-RI agents, bridging the RI disparity between water and other high-RI components (such as lipids and proteins) in endogenous tissues. However, our theoretical analysis shows that these molecules derive their high RI from absorption in the short-wavelength, EUV spectrum (Fig. 2). Equation 3 predicts that these molecules are weak Lorentz oscillators, characterized by an extremely small β (Table 1). This means that they inefficiently raise the RI of water per unit concentration when dissolved. This insight elucidates why these OCAs are typically used at very high concentrations [for example, 80% glycerol (*26*) or 80% fructose (*29*)] or in their pure forms [for example, pure glycerol (*27*)]. Consequently, conventional tissue clearing necessitates almost complete substitution of the original water content with the OCA, leading to substantial dehydration and shrinkage of the biological tissue (*2*, *5*). Such requirements make these traditional ex vivo tissue-clearing methods unsuitable for use in live biological tissues.

Validation of the mechanism enabling optical transparency with dye molecules

Increasing the absorption of a scattering medium has been shown to improve imaging resolution by suppressing scattered photons that contribute to the background noise (*12*, *13*). Our results, on the contrary, are not due to the absorption of scattered photons by dye molecules but originate from reduced scattering events in the tissue. To this end, we conducted experiments using tissue-mimicking scattering phantoms (Fig. 3A). We made these scattering phantoms by uniformly mixing silica spheres

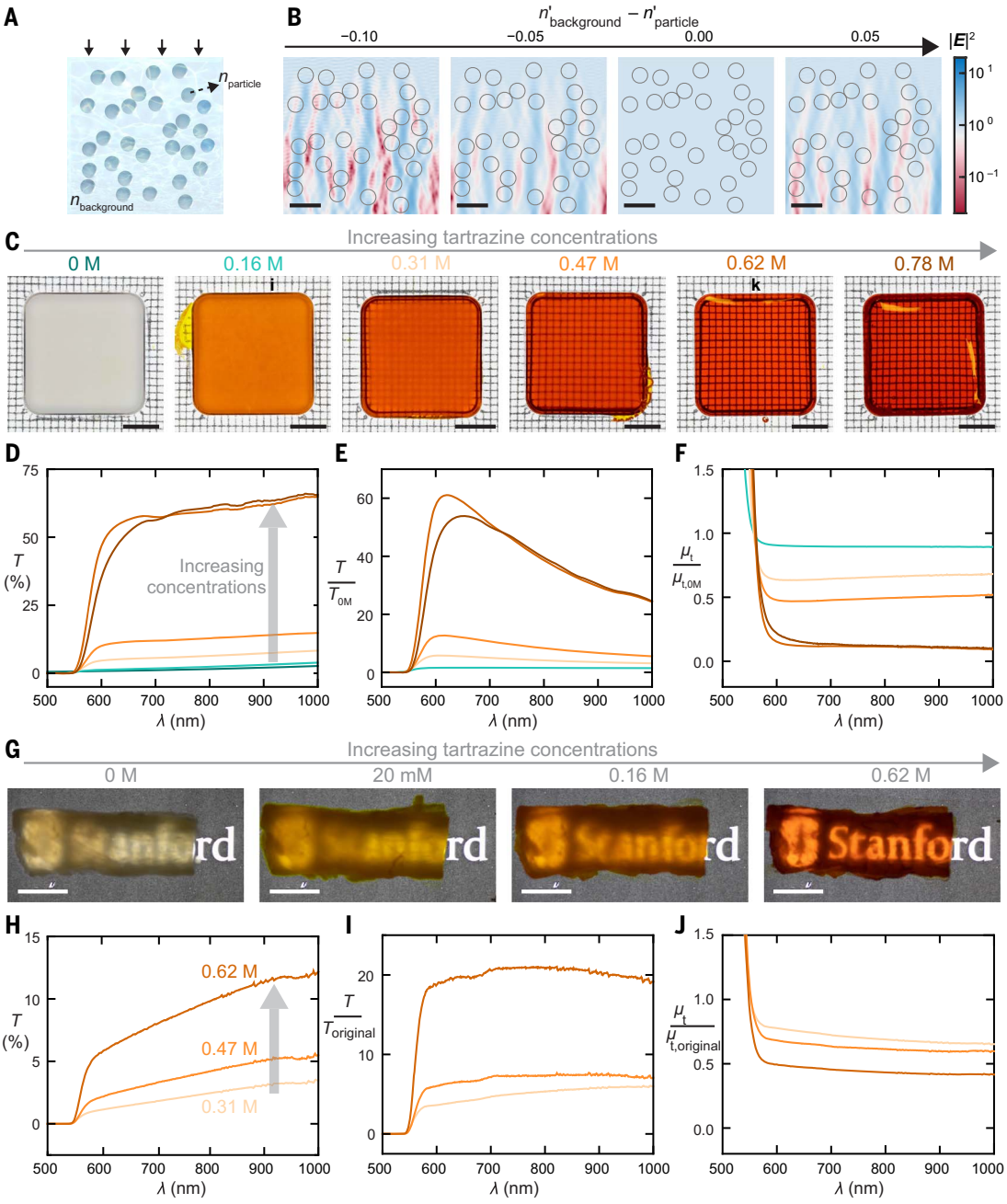


Fig. 3. Demonstration of optical transparency in scattering phantoms and ex vivo chicken breast tissue. (**A**) Schematic illustrating the scattering phantom composed of monodispersed silica particles dispersed in water. (**B**) Numerical simulation showing a plane wave traveling through a matrix comprising scattering particles in an aqueous background with different levels of n' mismatch indicated above each graph. Scale bars, 2 μm. (**C**) Photographs of scattering phantoms composed of agarose hydrogels containing increasing concentrations of tartrazine while keeping the concentration of silica particles the same. Scale bars, 5 mm. (**D**) Normal-incidence light transmittance T spectra of the scattering phantoms containing increasing concentrations of tartrazine. (**E**) Spectra of the transmission enhancement factor, which is defined as the transmittance ratio between the phantom containing a specific concentration of tartrazine and that containing 0 M tartrazine. (**F**) The ratio of attenuation coefficient between scattering phantoms containing different concentrations of tartrazine and that containing 0 M tartrazine. Curves in (D) to (F) follow the same color scheme as the concentration labels in (C). (**G**) Photographs illustrating the difference in the transparency of chicken breast tissue after soaking in tartrazine solutions with an increasing concentration. Scale bars, 1 cm. (**H**) Transmittance spectra and (**I**) transmission enhancement factor of the chicken breast tissue after immersion in different concentrations of tartrazine solutions. (**J**) Ratio of the attenuation coefficient of tissue immersed in tartrazine solutions of different concentrations over that of the original, uncleared tissue. Curves in (H) to (J) follow the same color scheme to indicate the concentration differences.

with an n' of 1.43 (close to that of lipids and collagen) in an optically transparent hydrogel with a background n' of 1.33 (the same as that of water). Agarose, used at a concentration of 6 mg ml^{-1} to solidify water into a hydrogel, had a minimal effect on the RI of the medium (fig. S17). We optimized the density of spheres to reach the same magnitude of the scattering coefficient of muscle tissue of ~10 cm^{-1} (fig. S18A) (*32*). Wave optics simulations based on the finite-difference time-domain (FDTD) technique show how the particles scatter light and disturb the propagation of the incident planar light waves (Fig. 3B). However, when we increased the n' of the hydrogel background to match that of the scatterers, the incident wave traveled without a noticeable distortion despite the presence of silica particles.

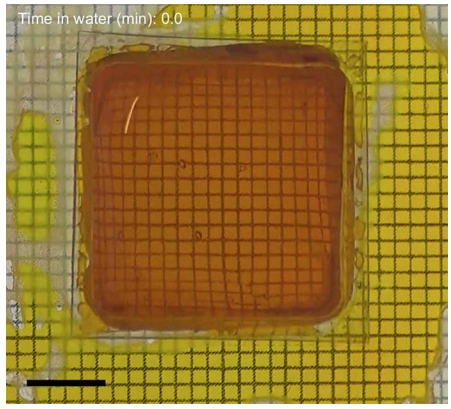
We placed the scattering phantom on top of a graphing paper showing a regular grid of millimeter-sized squares. Upon increasing the absorbance of this phantom by dissolving tartrazine into the hydrogel, the opaque phantom gradually turned transparent in the red part of the spectrum ($\lambda \geq 600$ nm) and appeared visually similar to the hydrogel without any scatterers, as evidenced by the brightfield photographs (Fig. 3C and fig. S19). Dynamic imaging of a scattering phantom immersed in a 0.6 M tartrazine solution shows it becoming increasingly transparent as tartrazine molecules gradually penetrate it (Movie 3), followed by a return to opacity as the molecules exit (Movie 4). We quantified the effect of optical transparency by measuring the normal-incidence light transmittance (T) through the phantom as a function of dye concentration (Fig. 3D). In the red transparency window ($\lambda \geq 600$ nm), we achieved a greater than 60-fold gradual increase of T with an increasing tartrazine concentration (Fig. 3E). At this concentration, the n' of the dye-doped hydrogel matches that of the silica spheres while producing a negligible n'' in the ≥600 nm range. Further increases in the dye concentration led to a decrease in maximum transmittance, which we attributed to an overshoot in n' that causes an increase in scattering, which agrees with the simulation results in Fig. 3B. These observations differ from the enhancement in image resolution induced by absorption; the latter displays a decrease in overall transmittance without exhibiting the overshoot behavior (*12*, *13*). The T measurements enabled us to extract the scattering coefficient μ_s using the Bouguer-Beer-Lambert law

$$T = \exp(-\mu_t d) \simeq \exp(-\mu_s d) \qquad (4)$$

where scattering provides the dominant contribution to the attenuation coefficient μ_t ($\mu_t = \mu_a + \mu_s \approx \mu_s$ because $\mu_s \gg \mu_a$) for wavelengths beyond 600 nm (fig. S18, B and C). We plotted normalized μ_t of the scattering phantom composed of increasing concentrations of tartrazine (Fig. 3F and fig. S18). We found a substantial reduction in $\mu_t/\mu_{t,0M}$ for scattering phantoms with a high concentration of tartrazine beyond 600 nm, indicating a reduction in light scattering as a result of n' matching. Using darkfield microscopy, we verified that the observed reduction in scattering is not due to the aggregation of silica scatterers (fig. S20). Moreover, we plotted $\mu_t/\mu_{t,0M}$ at varying wavelengths as a function of the n' of the aqueous background of the scattering phantom, which was doped with different dye concentrations (fig. S21). These plots exhibit good agreement with a reported monodisperse scattering model (*2*), thus confirming the RI modification by water-soluble dye molecules through the Kramers-Kronig relations as the origin of improved transmittance of the phantoms beyond 600 nm.

We validated the mechanism of optical transparency achieved by absorbing dyes using real biological tissue, such as chicken breast, as the scattering medium. The chicken breast has a μ_t of 32.5 ± 3.5 cm^{-1} at 600 nm (fig. S22), which is similar to the reported μ_s of ~30 cm^{-1} at 600 to 700 nm (*32*). Slices of chicken breast tissue (2 mm thick) soaked in solutions with increasing concentrations of tartrazine revealed an increasingly crisp image of the word "Stanford" that was positioned underneath (Fig. 3G). Similar to the demonstration in scattering phantoms, dynamic imaging revealed that the chicken breast tissue becomes increasingly transparent as it is gradually penetrated by tartrazine molecules (Movie 5). In the transparency window in the red wavelengths ($\lambda \geq 600$ nm), we achieved a gradual increase in T with an increasing concentration of tartrazine (Fig. 3, H and I). Moreover, we found a substantial reduction in $\mu_t/\mu_{t,original}$ for the chicken breast with an increasing concentration of tartrazine beyond 600 nm, indicating a reduction in light scattering as a result of n' matching rather than the absorption of scattered photons (Fig. 3J). A representative micrograph of the muscle section shows that tartrazine molecules penetrate myofibers and fill the aqueous sarcoplasmic space between the myofibrils (fig. S23). Consequently, tartrazine increases the RI of the sarcoplasm (1.35 to 1.37) to match that of the myofibrils (1.39 to ~1.47), resulting in the optical transparency of the muscle tissue.

Last, we measured the diffusion rates of tartrazine through common tissue types to evaluate its potential as a generic OCA. It has been reported that the dynamics of fluid diffusion within biological tissue can be approximated by means of free diffusion (*33*). By tracking the temporal evolution of optical transmission during dye immersion, we extracted the diffusion coefficients of tartrazine molecules in the range of 0.067×10^{-6} to ~0.66×10^{-6} cm^2 s^{-1} in various tissues—including the mouse skin, mouse muscle, rat skin, and chicken breast (fig. S24)—using a reported method (*34*). This range of diffusion coefficient is higher than that of other commonly used OCAs (for example, propylene glycol has been reported to have a small diffusion coefficient of 0.021×10^{-6} cm^2 s^{-1}) (*2*). Consequently, tartrazine takes much shorter time to achieve optical transparency (for example, 2 hours for the 2-mm chicken breast tissue) than the required immersion time for other OCAs [for example, 24 hours for 60% sucrose (*28*)] to achieve the same effect. We attribute this difference in diffusion rate to the much higher molar $\Delta n'$ coefficient β of tartrazine than that of conventional OCAs (Table 1). As a result, tartrazine can achieve the desired $\Delta n'$ at a much lower concentration than can conventional OCAs, leading to a more dilute solution with lower viscosity, which enhances its diffusion kinetics (*35*). Dynamic optical transmission measurements reveal that these tissues regain their original opacity as tartrazine molecules diffuse out, demonstrating the reversibility of the induced optical transparency (fig. S25).



Movie 4. Real-time imaging showing the dynamic process of a scattering phantom returning from transparency to opacity after immersion in water. Scale bar, 5 mm.



Movie 5. Real-time imaging showing the dynamic process of a chicken breast tissue transitioning from opacity to transparency after immersion in a 0.6 M solution of tartrazine. Scale bar, 5 mm.

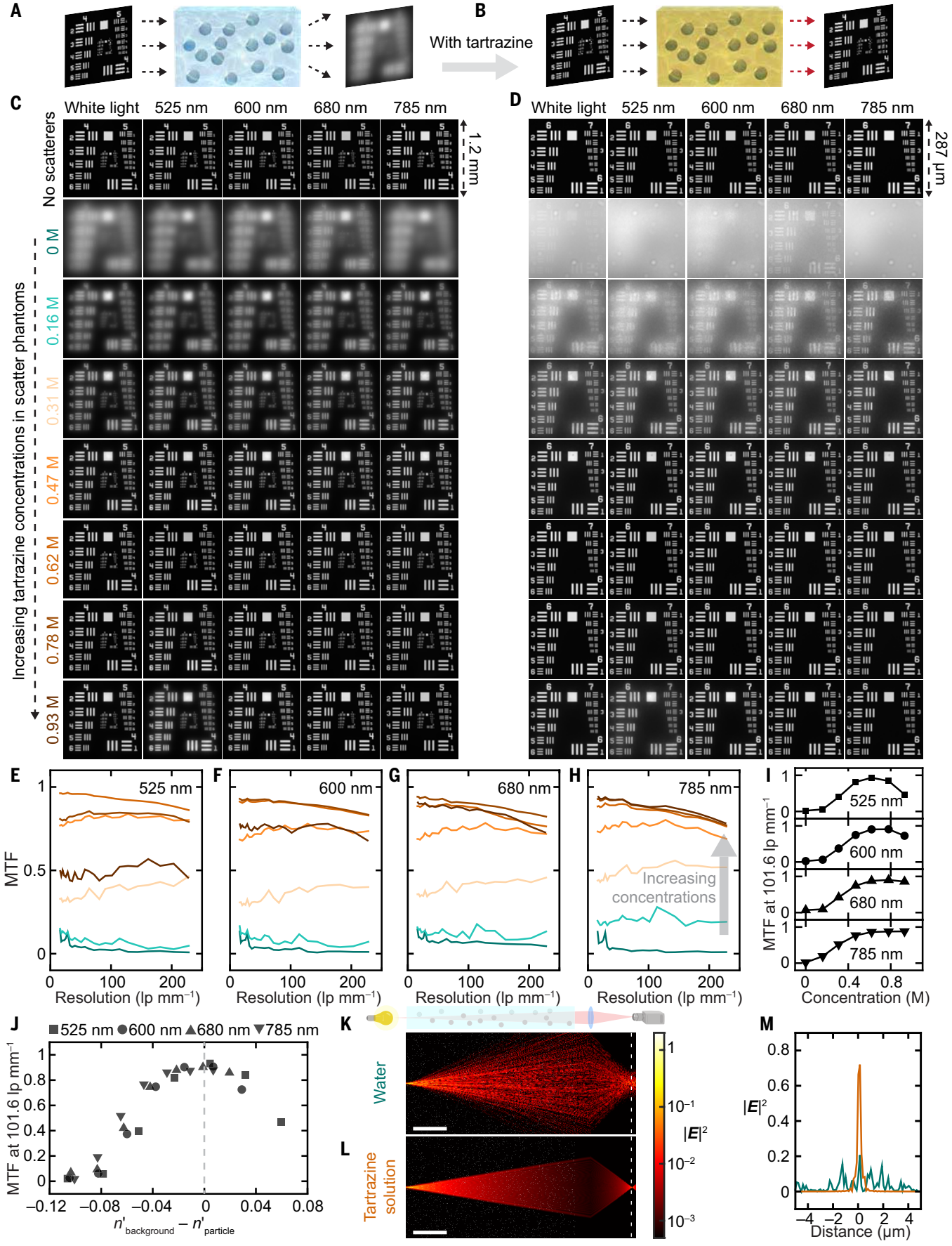

A
B
With tartrazine
C
White light
525 nm
600 nm
680 nm
785 nm
No scatterers
0 M
0.16 M
0.31 M
0.47 M
0.62 M
0.78 M
0.93 M
Increasing tartrazine concentrations in scatter phantoms
1.2 mm
D
287 μm
E
F
G
H
I
MTF
Resolution (lp mm⁻¹)
Increasing concentrations
MTF at 101.6 lp mm⁻¹
Concentration (M)
J
■525 nm ●600 nm ▲680 nm ▼785 nm
MTF at 101.6 lp mm⁻¹
n'background − n'particle
K
L
Water
Tartrazine solution
|E|²
M
Distance (μm)

Fig. 4. Imaging a resolution target through scattering phantoms containing different tartrazine concentrations. (**A** and **B**) Schematics showing the clearing effect through modulating the background RI in the scattering phantom through absorbing molecules, thus yielding higher resolution for imaging the USAF resolution target. (**C** and **D**) Intensity-normalized (C) low-magnification and (D) high-magnification images of the USAF resolution target through 1-mm scattering phantoms comprising different dye concentrations and imaged at different wavelengths. (**E** to **H**) MTF of the imaging system with scattering phantoms at (E) 525 nm, (F) 600 nm, (G) 680 nm, and (H) 785 nm wavelengths. Colors of the solid lines in (E) to (H) follow the same color scheme of the concentration labels shown in (C). (**I**) Dependence of MTF at a fixed spatial frequency of 101.6 lp mm^{-1} on the dye concentrations at different wavelengths. (**J**) Dependence of MTF at a fixed spatial frequency of 101.6 lp mm^{-1} on the n' difference between the background and particles in the scattering phantom. Symbols denote different wavelengths used for MTF measurements. (**K** and **L**) Simulated electric field E distribution of a 600-nm Gaussian beam launched from the left and propagating for 1 mm in a medium comprising silica scatterers (white dots) in (K) water and (L) an aqueous solution of 0.6 M tartrazine. The divergent beam is refocused by an imaginary lens placed 100 μm after light exits the scattering medium. The white dashed line indicates the plane where the propagating wave is refocused. Scale bars, 200 μm. (**M**) Electric field strength at the focal plane of light [along white dashed lines in (K) and (L)]. Blue, water; orange, 0.6 M tartrazine.

Dye molecules can improve the resolution of microscopic imaging within a scattering medium

Dye molecules can improve the spatial resolution in imaging at substantial depths in tissue-mimicking scattering media. We quantified the resolvable spatial frequencies of a 1951 US Air Force (USAF) resolution target when imaging it through 1-mm scattering phantoms with different tartrazine concentrations (Fig. 4, A and B, and fig. S26) (preparation of the phantom is described in the section Materials and methods). We placed the scattering phantom between the objective lens and the USAF target, which was back-illuminated with a white lamp (fig. S27). Micrographs through the clear phantom without any silica particles revealed the smallest features with a line width of 2.19 μm (Fig. 4, C and D, top row). However, the imaging resolution was greatly affected in the presence of the scattering phantom (Fig. 4, C and D, second row). The spatial resolution progressively increased as we increased the dye concentration in the phantom (Fig. 4, C and D, bottom rows), until clear resolution of the 2.19-μm line width. We quantified this gradual increase in spatial resolution through a measured increase in the modulation transfer function (MTF) (*36*) across all considered spatial frequencies and wavelengths (Fig. 4, E to H). The MTF displayed a wavelength-dependent behavior. At shorter wavelengths, images became blurry once surpassing an optimum concentration, whereas at longer wavelengths, the impact on both MTF and spatial resolution reached a plateau (Fig. 4I).

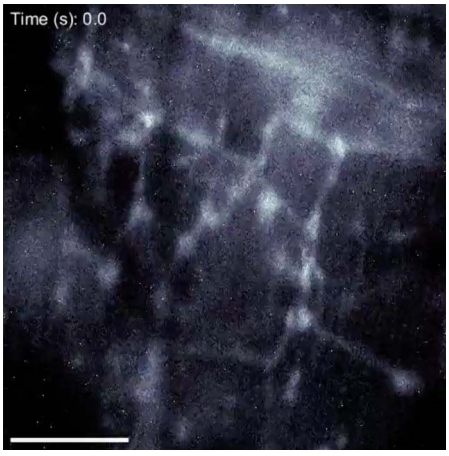
This wavelength-dependent behavior is consistent with the spectrally dependent changes in the RI as a result of dye doping. At shorter wavelengths, the real index of the aqueous (background) component $n'_{background}$ can overshoot the index of the silica particles. At longer wavelengths, the weaker dependence of the index enabled us to accurately dial in $n'_{background}$ to achieve transparency across a broad wavelength range (Fig. 2H). When we plotted the MTF at 101.6 line-pair per millimeter (lp mm^{-1}), which is relevant for imaging micrometer-sized structures, against the difference between the modulated background real index ($n'_{background}$) and the particle real index ($n'_{particle}$), the data at all wavelengths collapsed to the same master curve (Fig. 4J). This master curve, along with the transparency effect primarily in the red part of the spectrum ($\lambda \geq 600$ nm), validates RI modification by water-soluble dye molecules through the Kramers-Kronig relations. We analyzed the MTF at various depths in the scattering phantom and found that 0.6 M tartrazine can resolve micrometer-sized features even at a depth of 3 mm. By contrast, scattering phantoms with pure water and 0.6 M glycerol could not resolve these features beyond 1 mm (fig. S28, A and C). MTF analysis for chicken breast tissue of different thicknesses revealed comparable performance by 0.6 M tartrazine, resolving features as small as 5 μm at depths beyond 1 mm (fig. S28, B and D). Similarly, the USAF resolution target became resolvable underneath various mouse tissues in a tartrazine solution (fig. S29), consolidating the observed transparency effects in live mice.

Last, we carried out numerical, wave-optics simulations to model the interaction of an incident beam with scattering phantoms doped with and without tartrazine (Fig. 4, K to M). After the 600-nm light beam propagates through a 1-mm-thick scattering phantom, the beam is refocused by an ideal lens (Fig. 4, K and L); comparison of the point spread function highlights that a sharp focus can be achieved in the presence of tartrazine, in contrast to the original blurred focus in water (Fig. 4M). The simulations reveal the reduced scattering of 600-nm light waves by tartrazine owing to the RI modification following the Kramers-Kronig relations, rather than the removal of scattered photons by absorption, as the mechanism of the observed optical transparency. Taken together, these data demonstrate the ability to resolve spatial frequencies of up to 228.1 lp mm^{-1} (line widths down to 2.19 μm) through millimeters-thick scattering phantoms that optically mimics biological tissue.

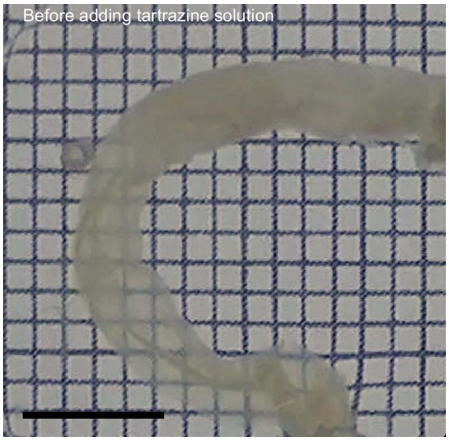
Monitoring gut motility through a transparent abdomen using dye molecules

We investigated whether the optical transparency would enable us to obtain high-resolution structural and functional information from deep-seated tissues that typically resist noninvasive microscopic imaging in live animals. We focused on the enteric nervous system (ENS), populating the gastrointestinal (GI) tract situated deep within the abdominal cavity. The ENS controls GI functions such as motility, luminal secretion, and absorption. Dysfunctions in the ENS are implicated in GI disorders such as irritable bowel syndrome and constipation, as well as the GI manifestations of neuropsychiatric illnesses such as autism and Parkinson's disease (*37*, *38*). Neuronal cell bodies of the ENS are grouped in clusters (ganglia) and are found within two mesh-like layers in the GI tract wall: the submucosal plexus and the myenteric plexus (*39*). The myenteric plexus, positioned between the circular and longitudinal muscle layers, contains local reflex microcircuits that sense intestinal distension and initiate peristalsis, the rostro-caudal propulsion of luminal contents. Despite the ENS's role, our understanding of how myenteric neurons orchestrate gut motility is limited. Direct imaging of myenteric neurons during gut motility in live animals is hindered by the opaque nature of the abdominal wall. Existing techniques to monitor gut motility often involve placing a dissected tissue in an ex vivo organ bath (*40*) or require the invasive surgical implantation of an abdominal window (*41*, *42*). The latter approach not only restricts normal gut movement but can also lead to colonic obstruction (*41*). These limitations underscore the need for noninvasive methods to image the movement of the GI tract and the ENS in their most native state in live animals.

We achieved optical transparency in the abdomen of live choline acetyltransferase (ChAT)–Cre-tdTomato mice. ChAT-Cre-tdTomato mice express tdTomato specifically in cholinergic neurons, which are excitatory within the ENS (*43*). We then performed fluorescence microscopic imaging of the myenteric plexus in the jejunum (Fig. 5A and fig. S30A), which is situated 500 to ~800 μm beneath the skin's surface (fig. S30G). In untreated animals, no discernible images could be obtained at this depth range owing to the scattering of overlying tissues (Fig. 5B and figs. S30B and S31, A to D). However, the dye molecules effectively rendered the skin and muscle above the jejunum transparent, allowing us to directly visualize the cholinergic neurons, along with



Movie 6. Real-time fluorescence imaging of the dynamic movement of the myenteric plexus through the transparent abdomen of a ChAT-Cre-tdTomato mouse whose cholinergic neurons in the gut are labeled with tdTomato. Scale bar, 100 μm.



Movie 7. Real-time imaging showing the dynamic process of a mouse small intestine tissue transitioning from opacity to transparency after immersion in a 0.6 M solution of tartrazine. Scale bar: 5 mm.

their nerve fibers with <10-μm diameters, labeled by the red fluorescent protein tdTomato in the myenteric plexus (Fig. 5C and figs. S30C and S31, E to M). We confirmed the fluorescence origin from jejunum neurons by dissecting the overlying skin, muscle, and jejunum for ex vivo imaging (fig. S30, D to H). Unlike the abdominal window, the "transparent abdomen" achieved by using dye molecules does not involve any exogenous rigid materials that restrict natural gut movement. Consequently, we were able to monitor the rapid dynamic movement of the myenteric plexus—reflecting the underlying motility of the jejunum—at a high frame rate of 28 frames per second (fps) (Fig. 5D and Movie 6).

We used particle image velocimetry (PIV), a technique commonly used to visualize the flow of fluids seeded with tracer particles (*44*), to analyze the dynamic movement of the myenteric plexus, using tdTomato-labeled myenteric neurons as "landmarks" to assist in the PIV analysis. Specifically, by conducting a PIV analysis on the video and excluding frames associated with global movements attributable to heartbeat and breathing, we generated a time-evolving vector map. In this map, each subregion is assigned a vector that signifies the orientation and magnitude of tissue movement (Fig. 5D). We first investigated the temporal evolution of the global movement pattern of the myenteric plexus by calculating the average moving direction and magnitude (Fig. 5E). This global movement pattern revealed a periodic change in the moving directions coupled with occasional bursts of large movement magnitude. To visualize such periodic movement, we selected a representative time series highlighting different moving directions and magnitudes (Fig. 5F). These time-evolving vector maps exhibited several informative findings. First, there exist periodic oscillations of global translation directions and switching between relatively featureless low-displacement maps and those with obvious movements (Fig. 5F). Second, various subregions within the field of view display patterns of tissue movement that deviate from unidirectional translation (Fig. 5G). For example, we observed contraction (Fig. 5H), expansion (Fig. 5I), and rotation (Fig. 5J) in representative examples. To highlight these local movement patterns, we computed the divergence of the vector map to pinpoint regions that act as "drains" and "sources" for tissue movement in contraction (Fig. 5H) and expansion (Fig. 5I). Similarly, we also computed the curl of the vector map for visualizing the rotation center (Fig. 5J). Third, the diversity of these movement patterns, all coexisting in a larger field of view at the same time, effectively rules out the possibility that they are solely due to the mouse's breathing and heartbeat. These physiological processes would typically induce a global, vertical translational motion across the entire field of view. Our approach offers the capability to visualize the activity of deep-seated tissues in live animals in their most native state.

Discussion

We report a counterintuitive finding: Strongly absorbing dye molecules, when applied topically to biological tissues, can reduce the intrinsic light scattering within these tissues. This effect renders various biological tissues—including the scalp, muscle, and abdomen—transparent. We constructed a theoretical framework based on the Lorentz oscillator model to explain this phenomenon. This model predicts that molecules with absorption at low resonance frequencies (long wavelengths) are more effective at raising the RI of water when dispersed than those with resonances at higher frequencies. To corroborate this hypothesis, we conducted extensive experiments and analyses using both tissue-mimicking scattering hydrogels and ex vivo biological tissues. These tests confirmed the mechanism behind our observations and showcased the achievable spatial resolution through millimeters of scattering medium once transparency is attained. This approach enabled us to transform the typically opaque abdomen and skin of a live mouse into a transparent medium in a reversible and repeatable manner (figs. S3, S6, S8, and S9). This "transparent abdomen" allows for direct visualization of fluorescent protein–labeled enteric neurons, capturing their movements that mirror the underlying gut motility in live mice.

Several limitations remain for this approach at its current stage. Specifically, this approach, like other similar in vivo tissue-clearing methods (*15*, *25*), is based on matching the RI of the aqueous media to that of the lipids and proteins. However, because the RI of these lipids and proteins is heterogeneous (ranging from 1.4 to 1.5) (*45*), it remains challenging to find a single, optimal concentration of the dye solution for perfect RI matching in a given tissue. Therefore, the observed optical transparency in ex vivo and in vivo tissues results from reduced, but not completely eliminated, scattering through them. In addition, because RI matching results from absorbing molecules dissolved in the aqueous medium, the achievable penetration depth depends on the diffusion of these molecules within a reasonable time frame (fig. S28). For thicker and less permeable tissues, the injection of absorbing molecules offers an alternative method for achieving tissue transparency at greater depths (fig. S7).

Our theoretical framework predicts the possibility of identifying even more efficient OCAs by searching for or synthesizing dye molecules that absorb in the NUV and blue regions of the spectrum (fig. S5). Although this may seem counterintuitive, such molecules function as Lorentz oscillators with more intense resonance strength, thus yielding a higher efficiency in matching the RI in biological tissue on a per molar basis. Contrasting this with conventional clearing agents, tartrazine is >15 times more efficient at RI matching in biological tissues, whereas Dye-21 is >100 times more efficient at achieving the RI matching in the NIR spectrum (Table 1). Consequently, tartrazine can achieve in vivo optical transparency at safe concentrations with acceptable osmolarity when applied topically. Molecules with lower resonance frequencies, sharper absorption peaks, and richer delocalized electrons are expected to be more efficient than tartrazine in achieving optical transparency in living tissues (supplementary text 2).

The safe concentration required for optical transparency is also seen in its compatibility

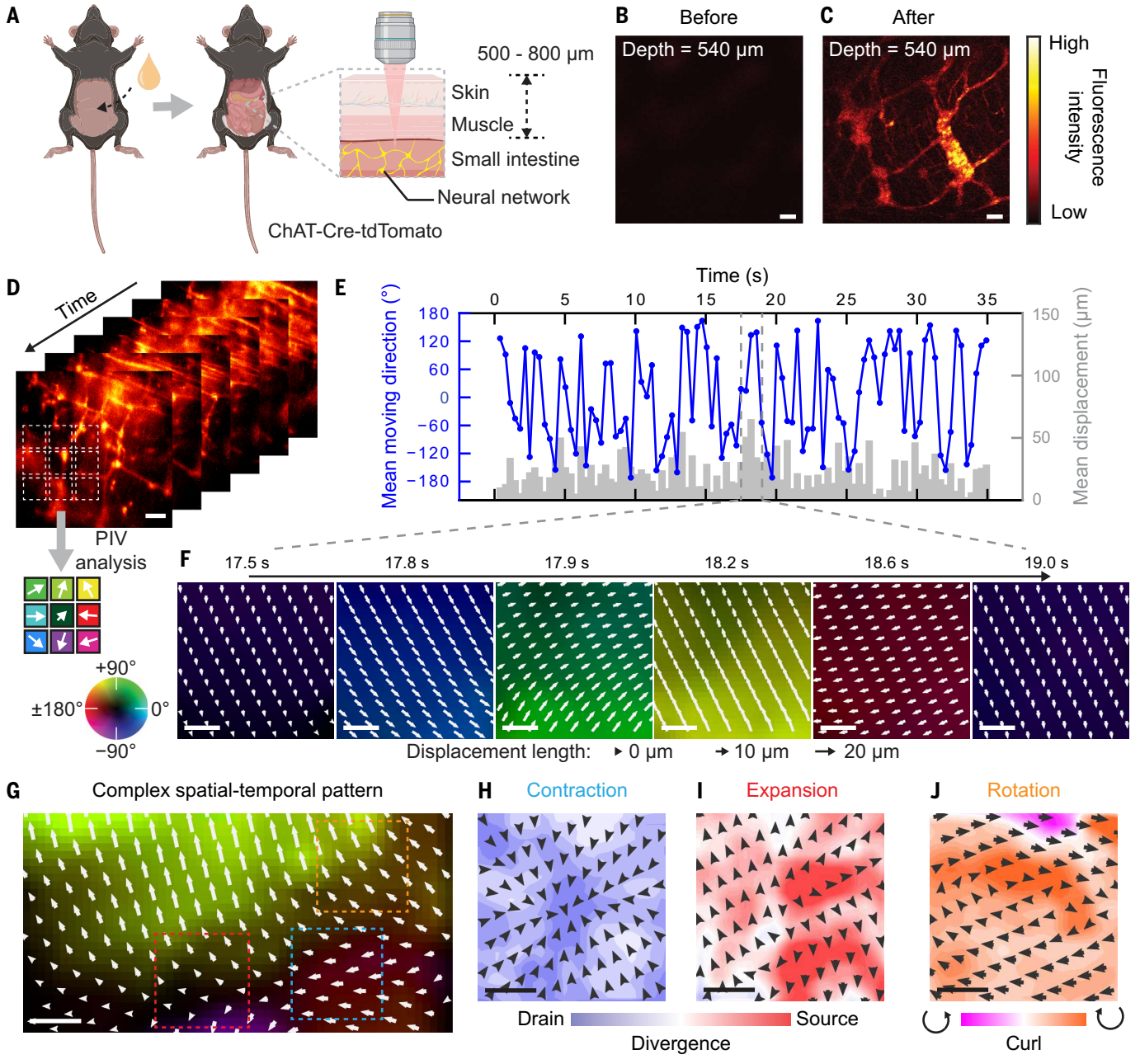


Fig. 5. High-resolution dynamic imaging of the mouse ENS through a transparent abdomen. (**A**) Schematics showing microscopic imaging through the transparent abdomen of mice. (**B** and **C**) Fluorescence images of the mouse myenteric plexus (B) before and (C) after optical transparency is achieved. (**D**) Widefield fluorescence image series of the mouse myenteric plexus, overlaid with the illustration of local displacement mapping. The direction and magnitude of movement are indicated with corresponding colors and arrows. (**E**) Temporal evolution of the average moving direction and displacement magnitude over intervals of 0.36 s in a 220- by 220-μm region. (**F**) Spatial-temporal evolution of plexus displacement between consecutive frames (with an interval of 0.03 s), with direction encoded in the background color and magnitude in vector lengths. (**G**) A snapshot highlighting the diverse patterns of local plexus movements in a representative frame. Dashed boxes highlight contraction (blue), expansion (red), and rotation (orange), respectively. (**H** to **J**) Three representative local movement patterns observed: (H) contraction, (I) expansion, and (J) rotation. The colormap denotes divergence in (H) and (I) and curl in (J), calculated from the displacement field, respectively. Scale bars, 50 μm. [Schematics were created with BioRender.com]

with fluorescent proteins. As an example, tdTomato expressed in the myenteric plexus of ChAT-Cre-tdTomato mice shows no reduction in fluorescence intensity before and after immersion in an effective tartrazine solution for jejunum tissue transparency (Movie 7 and fig. S32). Furthermore, tartrazine can achieve optical transparency in tissues both before (fig. S32) and after fixation (fig. S33). Both technical grade (90%) and analytical reagent grade (98.7%) tartrazine exhibit similar effects in achieving tissue transparency in their deprotonated form (fig. S34). Our pharmacokinetic studies revealed that systemically administered tartrazine is excreted from the mouse body through urine and feces within 24 hours (fig. S35). Skin histology indicates that topically applied tartrazine induces minimal

inflammation (fig. S36). Last, long-term studies evaluating the mice's body weights and blood panels revealed minimal systemic toxicity from topically applied tartrazine (figs. S37 to S39).

Materials and methods

Chemicals and materials

Dye molecules used in this study are listed with their full International Union of Pure and Applied Chemistry (IUPAC) names, chemical structures, and sources in Table 2. Two tartrazine products with different purities (T0388, technical grade, ≥85%; 03322, analytical reagent grade, >97%) from Sigma Aldrich were used in this study. Specifically, Batch #SHBQ2748 of Product T0388 was used and certified to have a purity of 90% (https://www.sigmaaldrich.com/certificates/sapfs/PROD/sap/certificate_pdfs/COFA/Q14/T0388-BULKSHBQ2748.pdf). In addition, Batch #BCCL1683 of Product 03322 was used and certified by high-performance liquid chromatography (HPLC) to have a purity of 98.7% (https://www.sigmaaldrich.com/certificates/sapfs/PROD/sap/certificate_pdfs/COFA/Q14/03322-BULKBCCL1683.pdf). Figure S34 confirms that both products exhibit the same absorption profiles, with a slight difference in absorption peak intensity corresponding to their purity differences. The impurities in T0388 do not contribute to any absorption. Both products have demonstrated their ability to achieve optical transparency in live mice when in the deprotonated form. Hydrochloric acid (HCl, 36.5 to 38.0 w/w %) and nitric acid (HNO_3, 68.0 to 70.0 w/w %) were purchased from Fisher Scientific Company (Pittsburgh, Pennsylvania, USA) and used without further purification. The low melting temperature SeaPlaque agarose was purchased from Lonza (Basel, Switzerland). Silicon dioxide spheres (powder; 637238) were purchased from Sigma-Aldrich (Burlington, Massachusetts, USA). Silica sphere suspension (SISN1000) was purchased from nanoComposix (San Diego, California, USA). The 1X phosphate-buffered saline (PBS) was purchased from Gibco (Waltham, Massachusetts, USA). Water used in this work was purified with a Millipore Milli-Q Integral 10 water purification system.

Mass spectrometry

Tartrazine was dissolved with water to a concentration of 0.1 mg ml^{-1} and this solution was analyzed by means of liquid chromatography–electrospray ionization mass spectrometry (LC-ESI/MS) on the Waters Acquity UPLC and Thermo Exploris 240 BioPharma orbitrap mass spectrometer. Spectra were collected in positive full scan mode, mass range of 150 to 2000 Da, Orbitrap resolution 120000. The column was a Phenomenex Kinetex F5 2.6u 100A 150 by 2.1 mm with mobile phases 0.1% formic acid in water and 0.1% formic acid in acetonitrile in a gradient mode at a flow rate of 0.25 ml min^{-1}. The injection volume was 3 μl.

Preparation and imaging of hydrogel–silica scattering phantoms

The as-received silica particles were characterized with an Apreo S LoVac scanning electron microscope (Thermo Fisher Scientific, Waltham, Massachusetts, USA) and NanoBrook Omni particle size and zeta potential analyzer (Brookhaven Instrument, Holtsville, New York, USA) before use. The hydrogel–silica scatter phantom (Fig. 3) was prepared by first mixing the low-melting-temperature agarose and silica particles with water or tartrazine solutions of different concentrations. The final concentration of agarose and silica particles in the mixture was 6 and 10 mg ml^{-1}, respectively. The mixture was then heated in an oven at 70°C for at least 10 min to facilitate the dissolution of agarose. A 1-mL suspension was pipetted into a plastic base mold (15 by 15 by 5 mm; 22-363-553, Fisher Scientific). The solution was refrigerated at 4°C for 10 min for solidification. The scattering phantom was then placed on a home-printed transparent pattern illuminated by a white light-emitting diode (LED) panel for imaging.

The hydrogel–silica scattering phantoms (Fig. 4 and fig. S28) were prepared by mixing the silica particle suspension (1 μm, 10 mg ml^{-1}) with low-melting-temperature agarose and the OCAs at specific concentrations. For Fig. 4, the phantoms were prepared with tartrazine at different concentrations similar to Fig. 3. For fig. S28, the phantoms were prepared with glycerol or tartrazine, both at 0.6 M. The final concentration of agarose was 6 mg ml^{-1} for all phantoms. The mixture was then heated in an oven at 70°C for at least 10 min to facilitate the dissolution of agarose. A suspension of the mixture was pipetted and injected to fill the well of a silicone isolator with varying depths (JTR20R-A2 series, Grace bio-labs, Bend, Oregon, USA) mounted on a coverslip (24 by 40 by 0.15 mm, 12-544-C, Fisher Scientific). The top of the solution was sealed by another coverslip (22 by 22 by 0.15 mm, 3306, Thermo Fisher Scientific, Waltham, Massachusetts, USA), and the phantom was left at room temperature for slow solidification. The phantoms were then placed on top of the negative 1951 USAF resolution test target (R3L3S1N, Thorlabs, Newton, New Jersey, USA) and imaged by using an optical microscope (DM2700M, Leica Microsystems, Wetzlar, Germany) with a 10X objective (NA: 0.25, N Plan Epi BD, Leica Microsystems). White-light illumination was generated by an arc lamp (66921, Newport, Irvine, California, USA). Filters selecting different wavelength ranges were inserted in the front of a scientific complementary metal-oxide semiconductor (CMOS) camera (C11440, ORCA-Flash4.0, Hamamatsu Photonics, Japan). Bandpass filters centered at 525 nm (FF01-525/45-25), and 785 nm (LD01-785/10-25) were purchased from Semrock (Rochester, New York, USA). Bandpass filters centered at 600 nm (FB600-10) and 680 nm (FB680-10) were purchased from Thorlabs.

Dark-field microscopy of silica particles

To visualize the spatial distribution of silica particles in the agarose gel, dark-field images were captured by using a Leica DM2700M microscope equipped with a dark-field objective (N PLAN EPI 10x/0.25 BD, Leica) and CMOS camera (ORCA-Flash4.0 LT3 Digital CMOS camera, Hamamatsu). Centroids of individual particles were tracked by using algorithms previously developed for analyzing digital micrographs of colloidal particles (*46*).

Calculation of the MTF

The MTF measures the ability of an imaging system to reproduce contrast for various spatial frequencies. It represents how the system captures the detailed structure of an image based on the spatial frequency of line or sine wave patterns, which are measured in line pairs per millimeter (lp mm^{-1}), as demonstrated by the 1951 USAF target. The MTF was determined by calculating the normalized difference between the maximum and minimum transmitted light intensities to assess contrast modulation

$$\mathrm{MTF} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \quad (5)$$

where $I_{\max}$ and $I_{\min}$ represent the maximum and minimum intensity of the bars in the USAF target (*36*). The MTF calculation in this work is specifically tailored for a set of spatial frequencies supported by the USAF target.

Preparation of ex vivo chicken breast tissue samples

Raw boneless and skinless chicken breast was bought from local sources and kept frozen at −20°C. Before experiments, the frozen, raw chicken breast was allowed to thaw at room temperature for 10 min. The chicken breast was then sliced to the desired thickness (0.5 to 5 mm) parallel to collagen fibers to ensure that each piece has fixed muscle fiber orientation. The thickness of the tissue was measured with a digital caliper (500-196-30, Mitutoyo) by sandwiching the tissue between two coverslips (22 by 22 by 0.15 mm; 3306, Thermo Fisher Scientific). (*47*) The ex vivo tissue sample was sequentially soaked in a 20 mM tartrazine solution for 20 min, a 163 mM tartrazine solution for 20 min, and a 0.62 M tartrazine solution for 2 hours to yield tissue samples of different tartrazine concentrations in Fig. 3. To facilitate the soaking of the sample in the tartrazine solution, the sample was placed on an orbital shaker (SHKE4000-7, Thermo Scientific) at 100 rotations per minute (rpm) and 40°C.

Ellipsometry measurements of the RI of dye solutions

A Horiba UVISEL ellipsometer was used to measure the real and imaginary parts of the

RI of solutions containing different concentrations of a specific dye molecule. Specifically, 5 ml of the solution was added to a shallow petri-dish (35 mm diameter by 10 mm height; Corning Falcon, New York, USA). The solution covered the entire bottom of the petri dish, creating a flat, reflective air-liquid interface. Reflected light was measured at an angle of 69.85°, and the RI of the solution was calculated with a single interface semi-infinite reflection model. For all measurements, a 0.2-s integration time was applied at each wavelength.

Measurements of the molar n′ *change,* β

The molar n' change, β, was defined and measured according to the following quantitative relationship between the n' and c (the concentration) of a certain molecule dissolved in an aqueous solution

$$n'(\lambda) = n'_{\mathrm{w}}(\lambda) + \beta(\lambda)c \qquad (6)$$

where n'_{w} is the real RI of pure water, c is the molar concentration (in mol liter^{-1}, or M) of the dissolved molecule, and β is its molar n' change (in M^{-1}). n', n'_{w}, and β all have wavelength dependence. According to Eq. 6, by measuring the $n'(\lambda)$ of the solutions with ellipsometry, one can obtain $\beta(\lambda)$ by fitting $n'(\lambda)$ with c to a linear relationship.

The relevance of β lies in its ability to guide us through the selection of appropriate dye molecules as OCAs and the optimization of their concentrations to reduce scattering in biological tissues. For imaging at a specific wavelength of interest, λ_0, one should select dye molecules with the highest $\beta(\lambda_0)$ and lowest $\alpha(\lambda_0)$. Because light scattering in most tissues (such as skin, muscle, and brain) is primarily due to a difference in the real RI between water (1.33) and lipids and proteins (1.4 to 1.5), the required concentration of the chosen dye molecules can be approximately determined by calculating $(1.45 - 1.33)/\beta(\lambda_0)$. This formula, however, merely provides a preliminary estimate because the RI of lipids and proteins varies; thus, a singular dye concentration that perfectly matches the RI for any tissue does not exist. Consequently, precise concentration determination requires experimental optimization.

UV-vis transmission spectroscopy

A Thermo Fisher Scientific Evolution 350 UV-vis spectrophotometer was used to measure the transmission and absorption spectra of dye solutions, silica scattering phantoms, and ex vivo tissues. For all measurements, a 0.2-s acquisition time was applied at each wavelength. The wavelength range was 190 to 1100 nm during all measurements.

For transmission measurements of dye solutions, a 1-mm optical path quartz cuvette (high precision, two polished sides; Orient Analytics, Yixing, China), a 0.1-mm optical path quartz cuvette, and a 0.01-mm optical path quartz cuvette (White Bear Photonics, White Bear Lake, Minnesota, USA), were used. Specifically, the cuvette was rinsed with aqua regia (mixture of 36.5 to 38.0 w/w % HCl and 68.0 to 70.0 w/w % HNO_3, with a volume ratio of 3:1) and water sequentially and then dried before use. A reference spectrum was acquired with water alone, thus correcting for any optical losses from intrinsic water absorption and reflection at the air-quartz interfaces.

For transmission measurements of scattering phantoms, the phantom with the mold was placed in the light path of the UV-vis spectrometer. A clear hydrogel without silica scatterers in the mold was used as the reference to correct for any attenuation from baseline absorption and reflection at the air-hydrogel interfaces.

For transmission measurements of ex vivo tissues, the tissue sample was first mounted on a microscope slide (precleaned, 12-550-15, Fisher Scientific), which was placed in the light path of the UV-vis spectrometer to ensure that the entire incident beam was blocked by the tissue. For dynamic UV-vis experiments, the samples were attached to one side of a 1-cm path cuvette holding 3 ml of the dye solution, and time-elapsed measurements were carried out with a time interval of 5 min until the transmission reached a plateau.

Numerical simulation of light propagation in water and scattering phantoms

The two-dimensional FDTD (Lumerical) method was used to simulate the electric (E) and magnetic (H) field of light propagating through an area of 10 by 10 μm (Fig. 3B) or 600 by 1000 μm (Fig. 4, K and L) that contains two components: silica scatterers and background medium (water or an aqueous solution of tartrazine). As shown in Fig. 3B, a plane wave was launched from the top of the simulation box. As shown in Fig. 4, K and L, a tight-focused Gaussian source with a NA of 0.25, matching the objective used in imaging, was constructed with 400 plane waves by using a thin-lens approximation. A virtual lens with a focal length of 200 μm was placed 100 μm away from the scattering medium to refocus the light, mimicking the formation of an image on a camera. For scattering phantoms, silica scatterers with a diameter of 1 μm were randomly placed in the simulation domain with a density of 0.3 μm^{-2} (Fig. 3B) and 0.003 μm^{-2} (Fig. 4, K and L). The RIs of water and silica scatterers were taken from the literature (*48*), and the RI of the tartrazine solution was taken from ellipsometry measurement (Fig. 2). The refractive indices of water and the aqueous solution of tartrazine are the same as their corresponding hydrogels containing 6 mg ml^{-1} agarose (fig. S17). The simulation time was set as 7500 fs, with a time stability factor of 0.9. Because of the large simulation domain in Fig. 4, K and L, mesh accuracy was set at 2 with 10 mesh points per wavelength to balance the simulation running time and accuracy, and a conformal variant of 0 was used for mesh setting. The boundary was set as five perfectly matching layers.

Vertebrate animal subjects

Adult (20 to 30 g) male and female C57BL/6J mice (for brain vascular imaging, 14 to 15 weeks old, Jackson Laboratory, Bar Harbor, Maine, USA), young (10 to 14 g) female BALB/cJ mice and female C57BL/6J (for muscle sarcomere imaging, 3 to 4 weeks old, Jackson Laboratory), young (10 to 15 g) female thy1-YFP-H and C57BL/6J mice (for brightfield imaging of abdominal organs, 3 to 4 weeks old, Jackson Laboratory), and adult (15 to 30 g) male and female ChAT-Cre-tdTomato mice (for fluorescence imaging of ENS, 7 to 12 weeks old) were the animal subjects used in this study. Mice were group-housed on a 12 hours:12 hours light:dark cycle in the Stanford University Veterinary Service Center (VSC) with food and water provided ad libitum as appropriate. All animal experiments conducted were approved by Stanford University's Administrative Panel on Laboratory Animal Care (APLAC) in accordance with Public Health Service Policy on Human Care of Laboratory Animals guidelines and were approved by the Institutional Animal Care and Use Committees of Stanford University.

Topical application of dye solutions on live mice

Mice were weighed and anesthetized through intraperitoneal injection of a mixture of 16 mg kg^{-1} ketamine (Dechra Veterinary Products, Overland Park, Kansas, USA) and 0.2 mg kg^{-1} dexdomitor (Dexmedesed; Dechra Veterinary Products) with insulin syringes outfitted with a 30G needle. The degree of anesthesia was verified with the toe pinch method. To maintain the body temperature and prevent hypothermia, a homeothermic blanket (Harvard Apparatus, Holliston, Massachusetts, USA) was set to 40°C and placed underneath the anesthetized mouse. Lubrication eye gel (GenTeal Tears, Alcon, Fort Worth, Texas, USA) was applied on both eyes of the mouse to moisturize the eye surface. Hair removal lotion (Nair, Church & Dwight, Ewing Township, New Jersey, USA) was used for depilation of the mouse abdomen, head, and hindlimbs.

For topical application, a gel containing the dye molecules was prepared by mixing the low melting temperature agarose with different absorbing molecule solutions (tartrazine sodium salt, 18 to 30 w/w %; 4-aminoantipyrine, 38 w/w %; sunset yellow sodium salt, 28 w/w %) in a 20-ml scintillation vial (FS74504-20, Fisher Scientific) to reach a final agarose concentration of 3 to 6 mg ml^{-1}. The mixture was then heated in an oven at 70°C for at least 10 min,

followed by refrigeration at 4°C for 10 min for solidification. For imaging the gut, brain, and sarcomeres, the gel was topically applied to the depilated abdomen, scalp, and the medial part of the hindlimb skin, respectively. The underlying tissues and organs became transparent to the naked eye at ~5 min after gentle massaging of the topically applied gel on the skin. For thicker and less permeable skin, a microneedle applicator or an insulin syringe with a 30G needle (fig. S7) was used to deliver the solution of absorbing molecules without agarose.

Bright field imaging of abdominal organs in live mice

The hair on the mouse's abdomen was removed with Nair (Church & Dwight, Ewing Township, New Jersey, USA) before the topical application of the gel containing absorbing molecules. In this study, both tartrazine sodium salt and 4-aminoantipyrine have demonstrated their ability to achieve abdominal transparency. After the abdominal tissue has sufficiently absorbed the dye molecules, a clear lubricating eye gel (GenTeal Tears, Alcon, Fort Worth, Texas, USA) was applied to the abdomen. Subsequently, the area was covered with a coverslip to prepare it for imaging. The mouse was then exposed to white light illumination (10,000 cd sr m^{-2}) during imaging. Brightfield images were captured by using a home-built imaging system equipped with a color charge-coupled device (CCD) camera (DCU224C, Thorlabs). The CCD camera has an array size of 1280 by 1024 and a sensitivity range from 400 to 700 nm, where the quantum efficiency is above 50%. The camera was outfitted with a Tamron AF 90mm f/2.8 Di SP AF/MF 1:1 Macro Lens to capture the mouse abdominal images. No filters were used for imaging. An exposure time of 50 ms was used for taking brightfield images, and a frame rate of 15 fps was used for recording videos (Movies 1 and 2) in the brightfield mode.

LSCI of brain vasculature in live mice

The depilated mouse head (with or without the application of dye molecules) was imaged through a laser speckle imaging system (RWD Life Science, Kent, Delaware, USA) under the sliding mode of imaging with an acquisition time of 5 ms per frame. An incident power density of 6.68 mW cm^{-2} was used for the 785-nm laser. Consecutive frames collected in 15 s were averaged to improve the signal-to-noise ratio of all images. All intensity profiles (fig. S4) are normalized between the maximum and minimum values taken from both curves of each plot. In this study, tartrazine sodium salt, 4-aminoantipyrine, and sunset yellow sodium salt have all demonstrated their ability to achieve scalp transparency.

SHG imaging of hindlimb sarcomeres in live mice

After topical application of tartrazine on the medial part of the hindlimb skin, the sarcomere features in the underlying hindlimb muscles were imaged at a depth of ~220 μm below the surface of the skin in live mice. Specifically, in vivo sarcomere imaging was performed on a laser-scanning microscope (Prairie) equipped with a wavelength-tunable titanium-sapphire laser (Mai Tai, Spectra-Physics). We used 1000- or 1040-nm illumination with an incident laser power of 0.66 mW, and the emission was band-pass filtered in the 500/10 or 525/50 channel, respectively. A microscopic objective (Olympus) with a NA of 0.3, 0.5, or 0.8 focused illumination onto the mice limb. Images were acquired at 512 by 512 pixels with 6-μs pixel dwell time and averaged over 64 consecutive frames. A band-pass filter selective for 1- to 5-μm periods was then applied in ImageJ, following a previously reported protocol for sarcomere imaging (*19*).

Fluorescence imaging of myenteric plexus in live mice

After the topical application of the absorbing gel to the abdomen of ChAT-Cre-tdTomato mice, the fluorescence signal from tdTomato was collected by using a laser-scanning two-photon fluorescence microscope (Prairie) (Fig. 5, B and C). The imaging was performed under 920-nm illumination with an incident laser power of 46.8 mW, and the emission was band-pass filtered at 595/50. An objective with a 0.3 NA from Olympus (PlanN) focused the illumination onto the mouse's abdomen. Images were acquired at a resolution of 512 by 512 pixels with a 6-μs pixel dwell time. Intensity profiles (fig. S31) were acquired along the lines indicated in images taken at different depths, with the signal-to-background ratio (SBR) computed as the ratio of the peak intensity to the average baseline signal from the background. For faster imaging speeds (Fig. 5D), an epifluorescence imaging system (Bergamo imaging platform, Thorlabs) was used. This system was operated under 561-nm illumination, and its emission was band-pass filtered at 607/70. A 0.5 NA objective from Thorlabs focused the illumination onto the mouse's abdomen for imaging. The small-NA objective ensures a larger depth of field, accommodating the motion of the GI tract along the z axis during dynamic microscopic imaging.

After the in vivo imaging experiment, the abdominal wall of the imaged subject was removed, and the tissues were dissected for ex vivo fluorescence imaging by using a Zeiss 980 confocal microscope (fig. S30). For the quantitative comparison of fluorescence intensity (fig. S32), Fiji software was used based on methods described in previous publications (*49*). Regions of interest (ROIs) measuring 150.81 by 150.81 μm were selected on the focal plane for each image. The region exhibiting fluorescence signals was identified by using the autothresholding function in Fiji, and the remaining portion of the image was designated as the background region. Fiji then separately measured the mean intensity for both the signal and background regions. The fluorescence intensity was subsequently calculated using the formula: fluorescence intensity = mean(signal) – mean(background).

Analysis of the dynamic movement of the myenteric plexus

A custom Matlab code was developed to process the results from the fluorescence imaging. Initially, out-of-focus images owing to global body movements were discarded. Subsequently, displacement vector maps between adjacent frames were computed by using the PIV method. For calculating the displacement at a specific location between two images, a window of 128 by 128 pixels (equivalent to 93.44 by 93.44 μm) was centered around the location of interest in the first image. Simultaneously, a search window of 256 by 256 pixels (equivalent to 186.88 by 186.88 μm) was centered around the same location in the subsequent image. A two-dimensional cross-correlation was computed by using Matlab's built-in function, normxcorr2. The local peak of this cross-correlation, identified by the findpeaks function in Matlab, was designated as the displacement vector. This procedure was iteratively applied to each region of interest throughout the entire field of view, resulting in the two-dimensional displacement map (Fig. 5, F to J).

Metabolic study for quantification of excretion of injected tartrazine

Three mice were individually housed in a metabolic cage after being intramuscularly injected with 50 μl of 0.6 M tartrazine solution prepared with 1x PBS. Feces and urine samples were collected in the following 48 hours. Specifically, tartrazine remains the same molecule in the urine, as evidenced by an identical spectrum in the urine, whereas a metabolized form of tartrazine is found in the feces. The concentration of unmetabolized Dye-4 in the urine was determined on the basis of the absorption maximum at 428 nm, measured with the UV-vis spectrometer (fig. S35). In addition, feces were collected, hand ground, and mixed with deionized water to a final concentration of 10 mg ml^{-1}. The solution was then kept rotating at 70°C for 15 hours (micro autoblot hybridization oven; Bellco Glass, Vineland, New Jersey, USA) and filtered through a 0.45-μm syringe filter (SLHVR33RS, Sigma-Aldrich) before the UV-vis measurement. The metabolites of tartrazine are absorbing in the 300 to 700 nm range, thus allowing us to use an integrated area under the curve (fig. S35) to quantify their concentration by assuming the linearity in Beer-Lambert law.

Replication

The sample size for each experiment was determined by power analysis to ensure statistical rigor for all comparisons

$$n = \left[\frac{(Z_{1-\alpha/2} + Z_{1-\beta})\sigma}{|\mu_1 - \mu_0|}\right]^2 \quad (7)$$

where $\alpha = 0.05$, $Z_{1-\alpha/2} = 1.96$, $\beta = 0.20$, $Z_{1-\beta} = 0.84$, μ_0 is the mean of the outcome variable for the control (without treatment), μ_1 is the mean of pilot test samples after tissue transparency is achieved, and σ is the standard deviation of the outcome variable (*50*).

Brightfield imaging of internal organs through the transparent abdominal wall was repeated in six mice. LSCI of brain vasculature was repeated in eight mice. SHG imaging of muscle sarcomeres was repeated in four mice. Two-photon fluorescence imaging and widefield fluorescence microscope imaging of myenteric plexus was repeated in four mice.

Statistics

Comparison between groups was evaluated with a paired sample *t* test (OriginPro) and one-way analysis of variance (ANOVA). $P < 0.05$ was considered statistically significant.

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ACKNOWLEDGMENTS

Ellipsometry characterizations were performed at the Stanford Nano Shared Facilities (SNSF), supported by the National Science Foundation under award ECCS-1542152. Any opinions, findings, and conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. In vivo two-photon imaging experiments were performed at Stanford Wu Tsai Neuroscience Microscopy Service, with help from G. Wang. We gratefully acknowledge M. Goodman and A. Baghdasaryan at Stanford University for useful discussions and assistance. **Funding:** This work was supported by National Institutes of Health grant 5R00AG056636 (G.H.); National Institutes of Health grant 1R34NS127103 (G.H.); National Institutes of Health grant R01NS126076 (G.H.); National Institutes of Health grant R01NS129834 (X.C.); National Institutes of Health grant R01DA059602 (X.C.); National Institutes of Health grant R01MH116904 (X.C.); National Institutes of Health grant R01DA045664 (X.C.); National Science Foundation CAREER award 2045120 (G.H.); National Science Foundation EAGER award

2217582 (G.H.); National Science Foundation Graduate Research Fellowships program 1656518 (N.J.R. and C.H.C.K.); Air Force Office of Scientific Research grant FA9550-21-1-0312 (M.L.B.); a Beckman Technology Development grant (J.A.K. and G.H.); a Rita Allen Foundation Scholars award (G.H.); the Focused Ultrasound Foundation (G.H.); the Spinal Muscular Atrophy Foundation (G.H.); the Pinetops Foundation (G.H.); the Bio-X Initiative of Stanford University (J.A.K. and G.H.); a Bio-X Graduate Student Fellowship (Y.-S.D. and N.J.R.); a Stanford Interdisciplinary Graduate Fellowship (H.C.); a Wu Tsai Neuroscience Institute seed grant (G.H.); the Wu Tsai Neuroscience Interdisciplinary Scholar program (Z.O.); the Wu Tsai Neuroscience NeuroTech Training program (C.H.C.K.); a Stanford Graduate Fellowship (X.W.); the Knight-Hennessy Scholarship program (K.B.); and the US Army Long Term Health Education and Training program (K.B.). **Author contributions:** Conceptualization: Z.O., N.J.R., M.L.B., and G.H. Materials absorption characterization: Z.O. and B.C. Ellipsometry characterization: Z.O. Silica phantom characterization: Z.O., X.W., and F.Y. FDTD simulation: Y.-S.D. and M.L.B. Ex vivo tissue characterization: Z.O. and S.W. In vivo animal experiments: Z.O., C.H.C.K., S.J., K.B., S.Z., E.L.S., H.C., W.Q., A.T., R.R., J.D., X.C., J.A.K., and G.H. Supervision: J.A.K., M.L.B., and G.H. Writing – original draft: Z.O., Y.-S.D., N.J.R., M.L.B., G.H. Writing – review and editing: Z.O., Y.-S.D., N.J.R., C.H.C.K., E.L.S., A.T., R.R., J.D., X.C., J.A.K., M.L.B., and G.H. **Competing interests:** G.H., Z.O., N.J.R., S.J., M.L.B., Y.-S.D., K.B., and S.Z. are inventors on patent application (WO2023122534A1) submitted by Stanford University that covers the principles of achieving optical transparency by applying the Kramers-Kronig relations. **Data and materials availability:** All data are available in the manuscript or in the supplementary materials. Aqueous solutions and hydrogels containing dye molecules are available from G.H. under a materials transfer agreement with Stanford University. The code for all simulations and data analyses are provided on Zenodo (*57*). **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adm6869
Supplementary Text
Figs. S1 to S39
References (*58*, *59*)
MDAR Reproducibility Checklist

Submitted 1 November 2023; accepted 12 July 2024
10.1126/science.adm6869